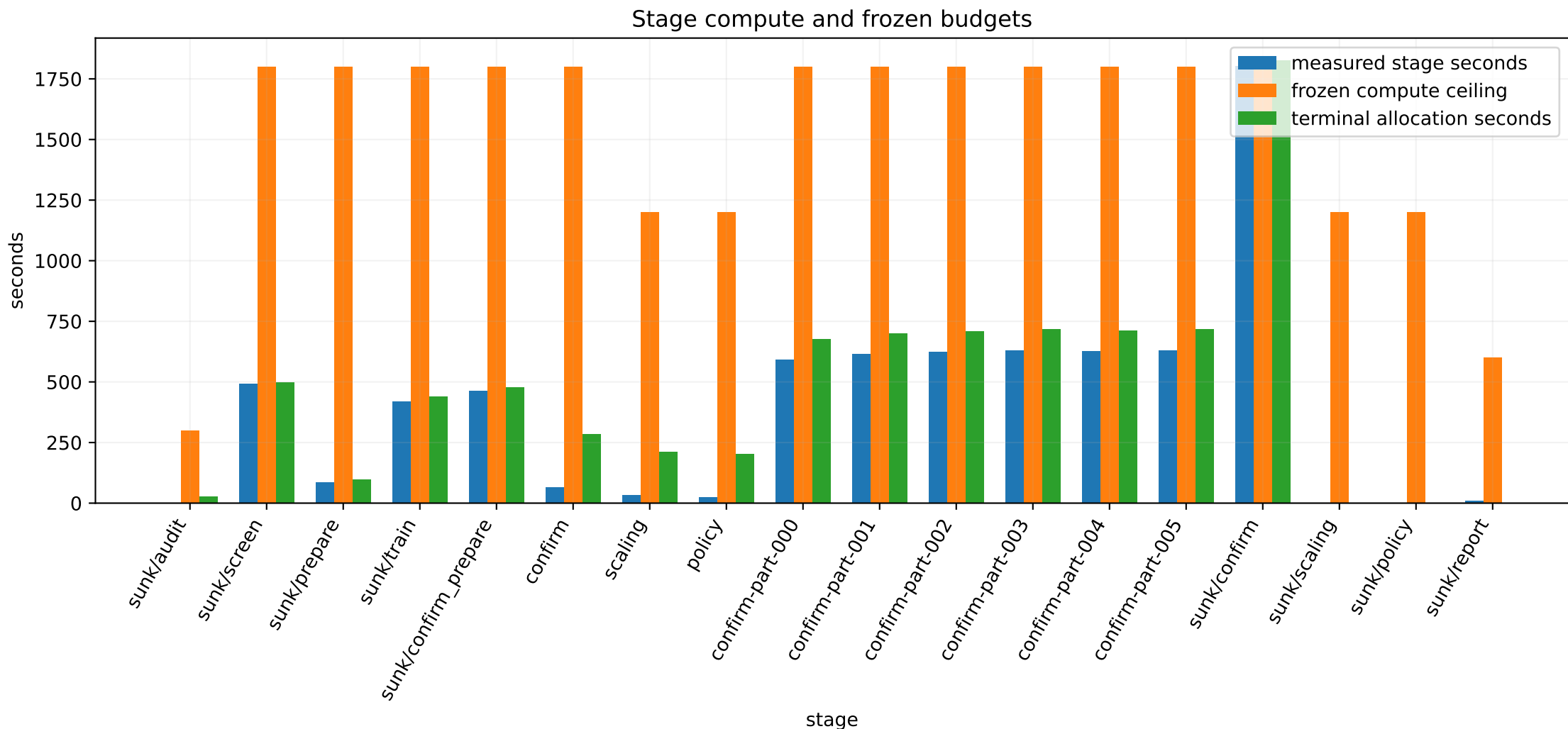


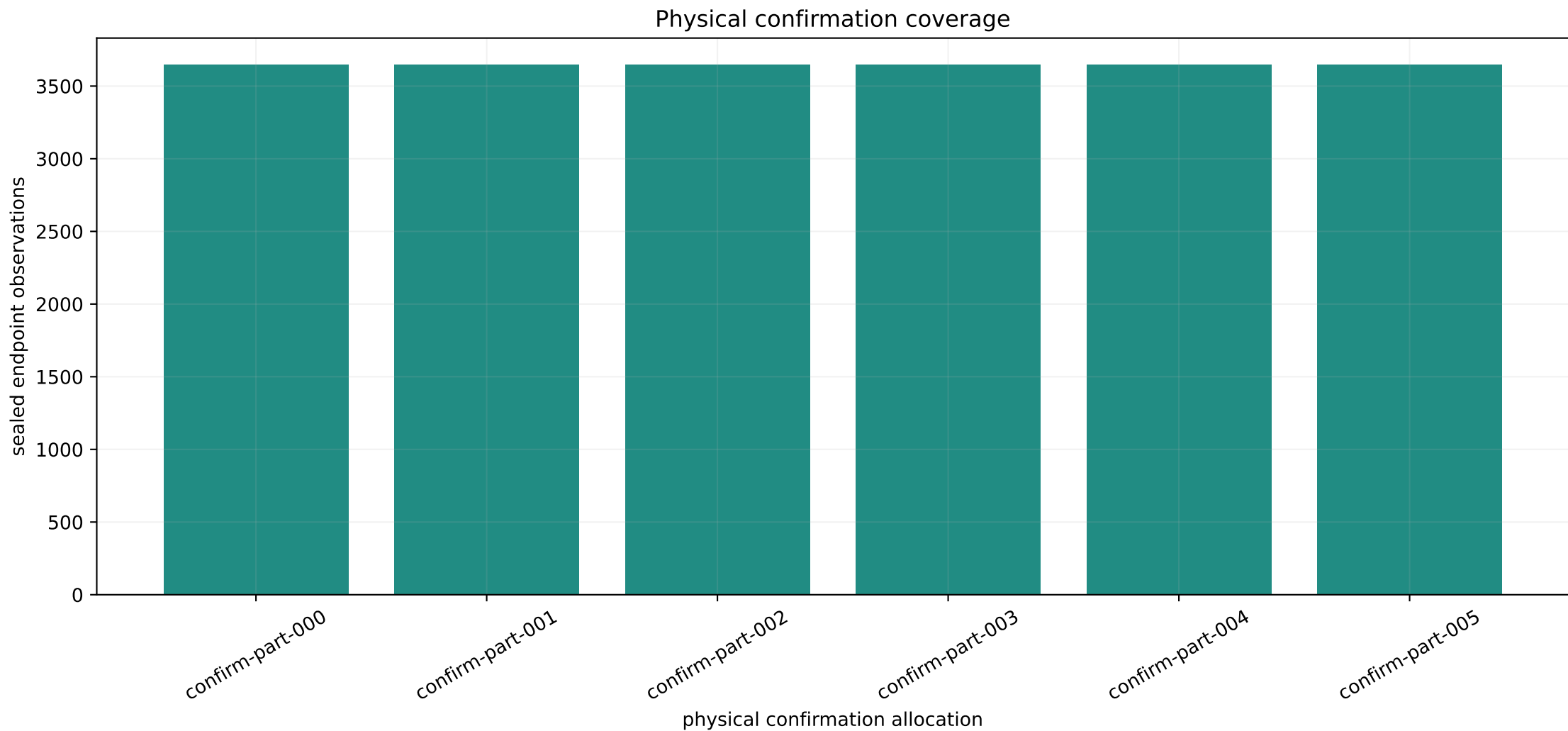


HOW TO READ: Y: lower measured/allocation seconds means less time for the same completed work. A budget is a ceiling, not consumed time; a fast failed stage is not a win. X lists stages, not a quality ranking.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Each physical allocation appears once; batch/step records are never added. Sunk/origin costs are separate from current work. Failed stage costs remain visible; running report allocation and tariffs remain NA.



HOW TO READ: Y: more sealed endpoints means more of the planned coverage, not greater accuracy. Full planned coverage is the goal; teal is verified, gray is unavailable. Repeated endpoints are not independent fields.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Partial partition observations are descriptive only. Missing or failed allocations remain NA; G3 requires the verified complete aggregate. Merged scientific rows count once.

Unmerged partial confirmation diagnostics

NA: no unmerged sealed partial endpoint errors

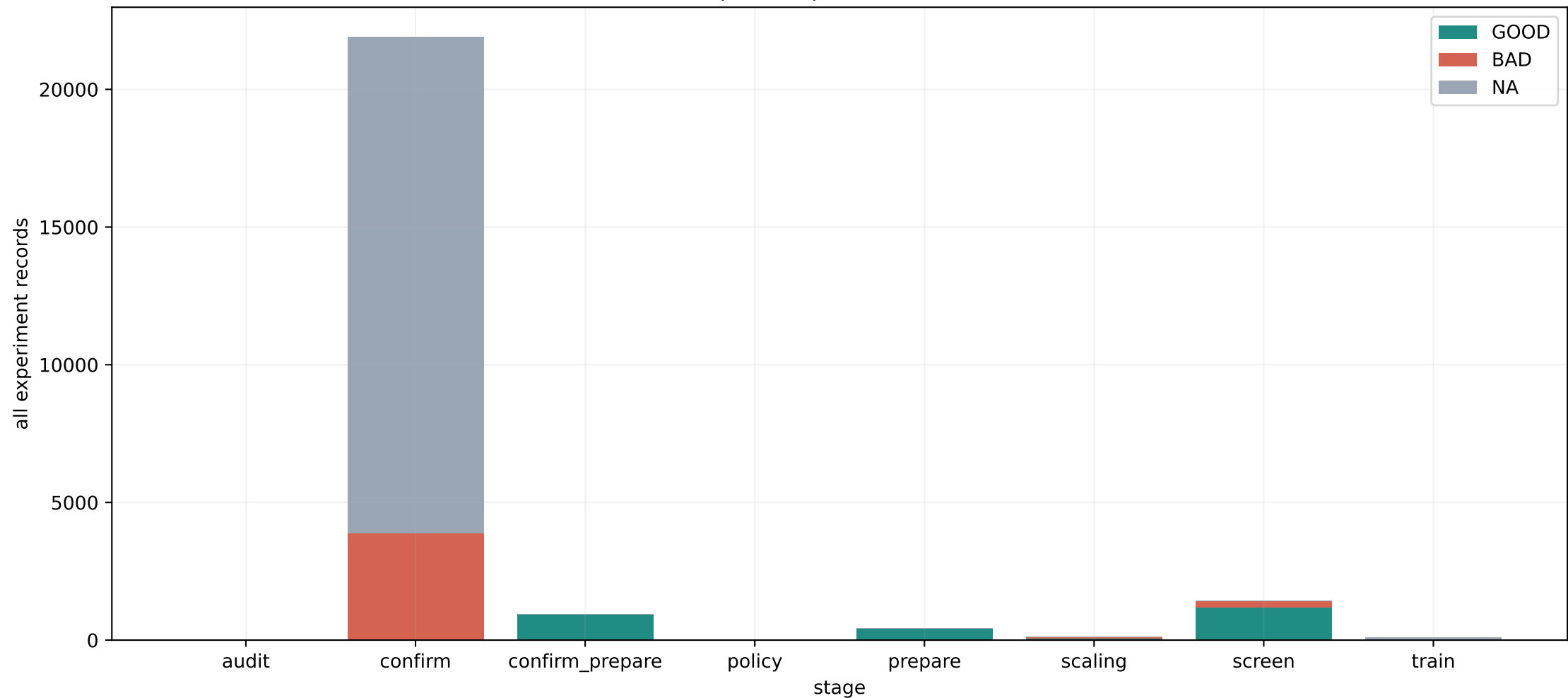
HOW TO READ: Y: lower RMS error is more accurate for matched cases. X groups physical parts. Blue is sealed partial evidence; gray is unsealed forensic evidence. Neither establishes a complete comparison; an empty panel is NA, not zero error.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Blue: sealed partial coverage. Gray: unsealed forensic groups (no scientific credit or resumption). All observations remain in chart-data.json.gz. This is not a model ranking, speedup estimate, or complete gate comparison.

Complete experiment verdicts

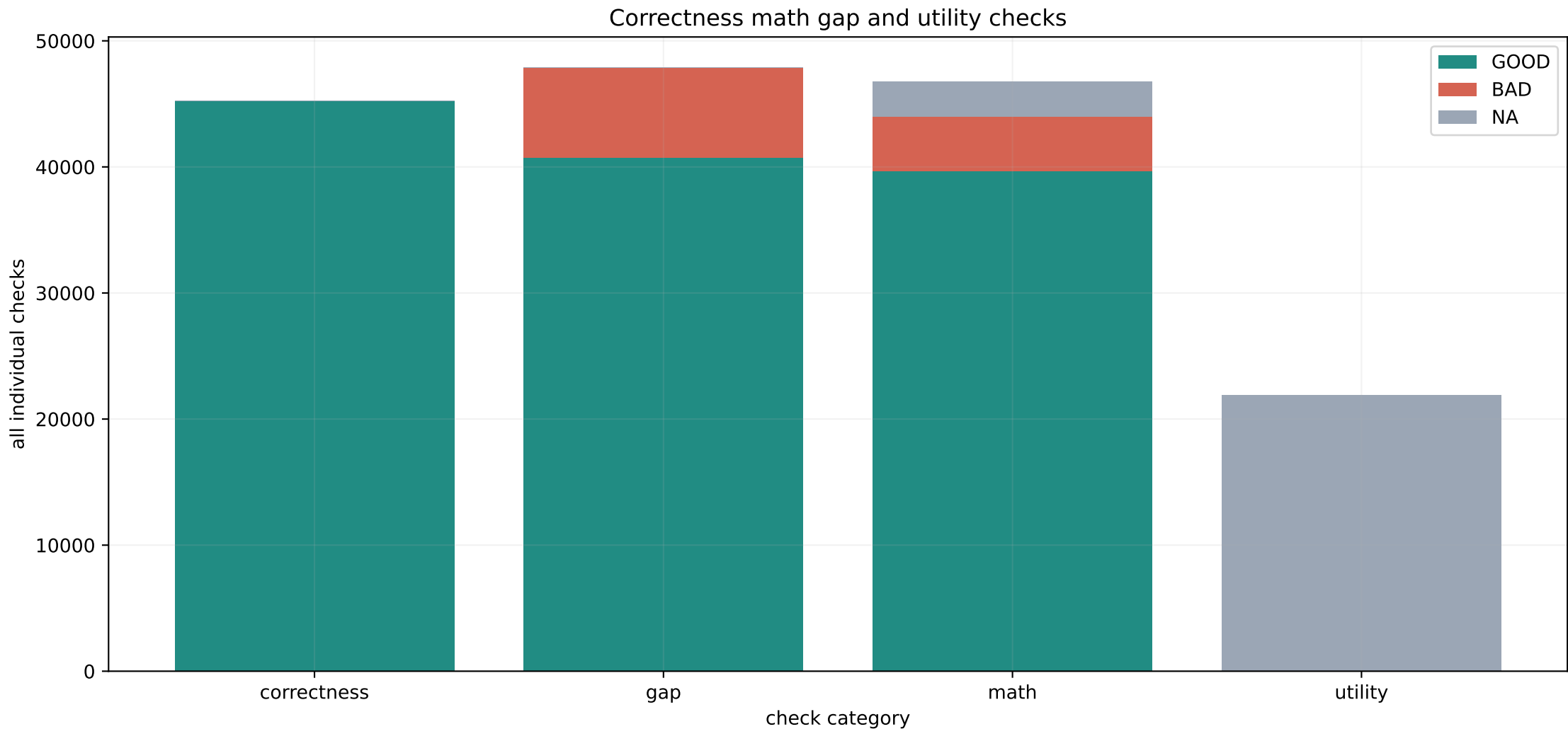


HOW TO READ: Teal GOOD means declared checks attained; coral BAD means failed requirements; gray NA means unresolved or unavailable. Y counts records, so taller bars do not mean better methods. Compare verdicts within the declared experiment scope.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: A required cost failure can make an otherwise accurate experiment BAD. Scores measure evidence attainment, not model quality.



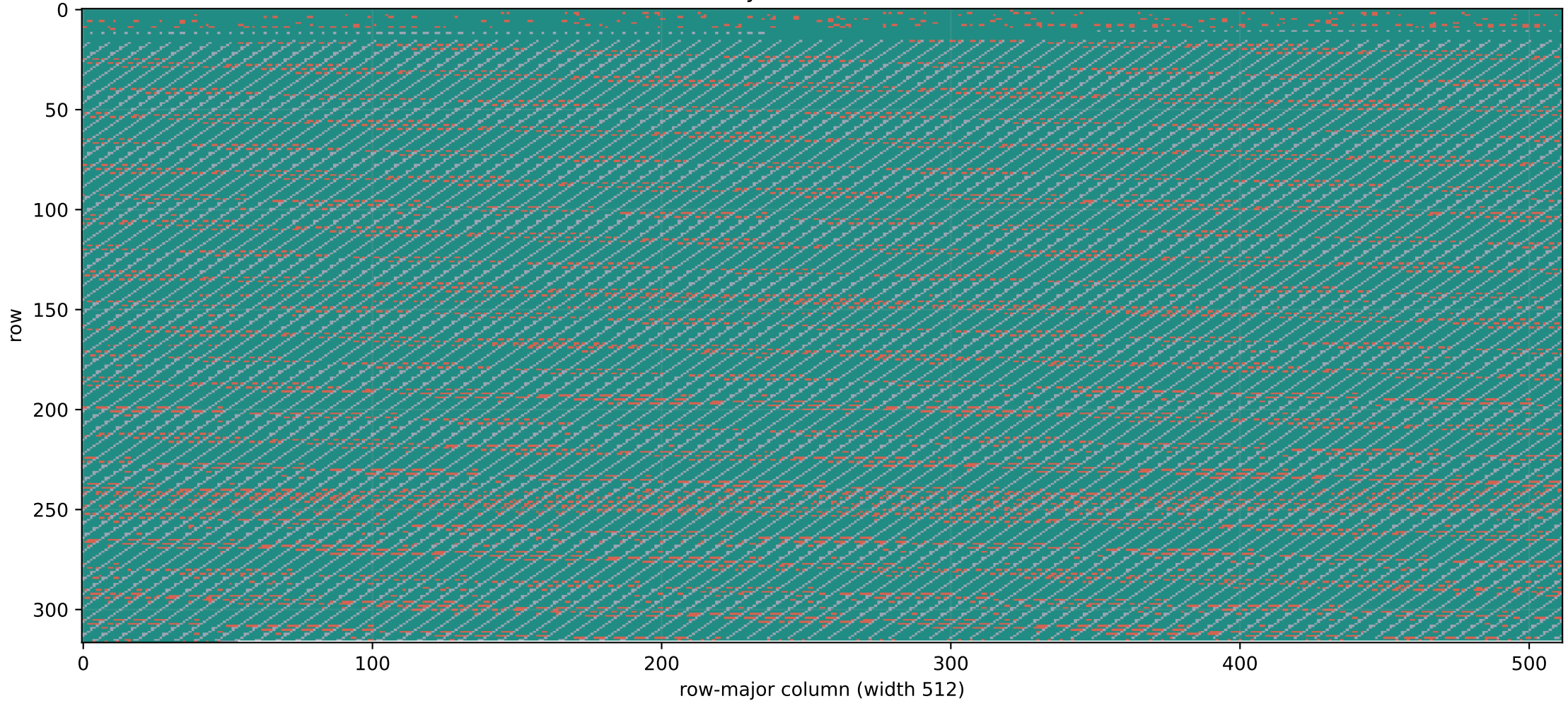
HOW TO READ: Teal GOOD is favorable check evidence; coral BAD is unfavorable; gray NA is unresolved. Y counts checks, not effect size or independent replications. More checks or passing numerical examples do not prove a mathematical theorem.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Numerical agreement is finite evidence, never a general proof. Missing evidence is retained as NA.

Every individual check



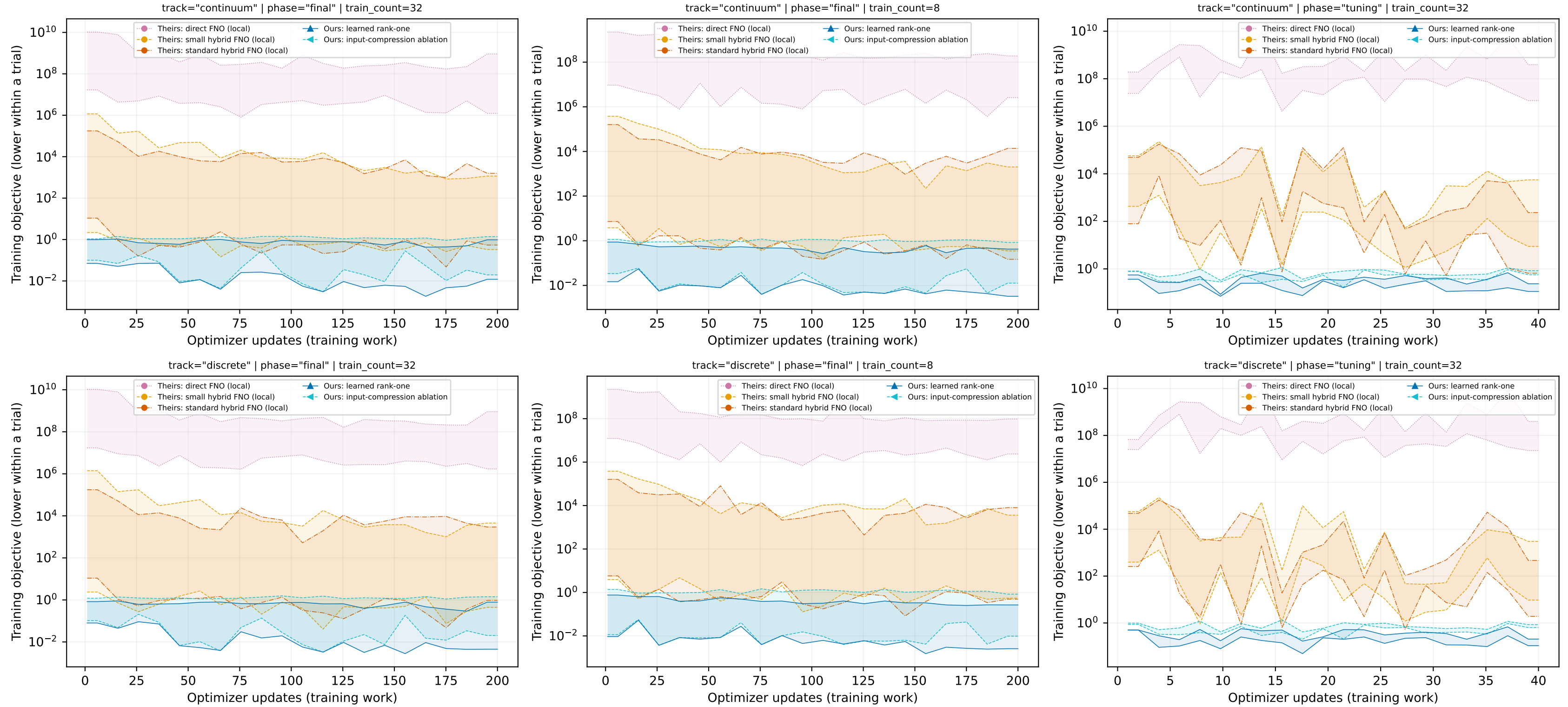
HOW TO READ: Color is the result: teal GOOD, coral BAD, gray NA; white is padding. X/Y locate checks in row-major order and have no better/worse direction. NA is not a pass; the lookup file identifies every cell.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: 161846 cells; no downsampling. Teal GOOD, coral BAD, gray NA, white padding. Exact identities and values in [check-cell-index.json.gz](#).

Training loss by update — observed low-high range



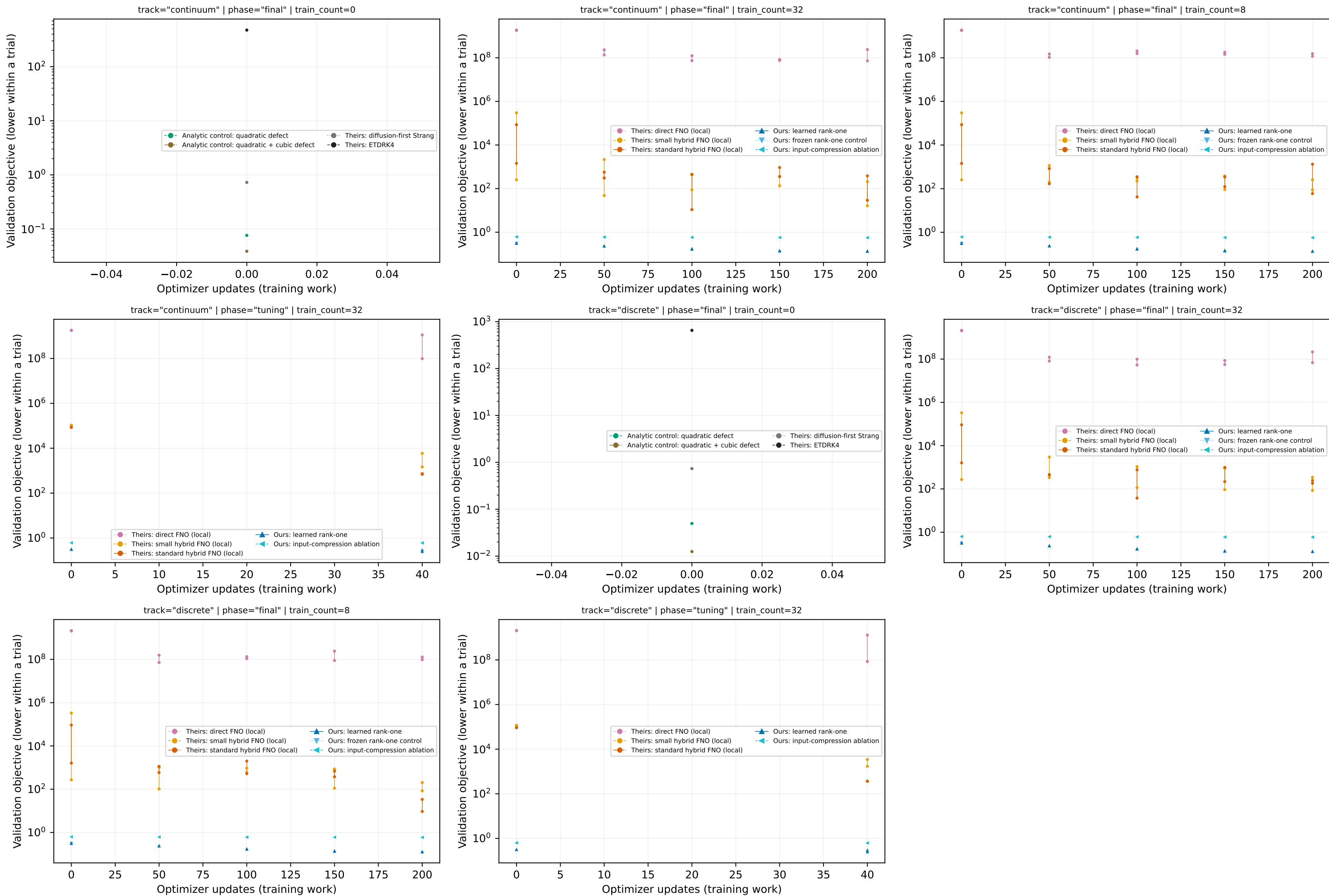
HOW TO READ: Y: lower training loss means a closer fit to that training objective. X: farther right means more optimizer updates, not inherently better quality. Compare like objectives; training loss alone does not establish generalization.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Observed min/max in up to 20 shared x-windows across available seeds/trials; tuning includes all settings. Continuous connections interpolate window summaries, not additional observations or confidence intervals. Phase/data budgets stay separate; empty windows and single-x checkpoints break bands. Exact bounds/counts: [learning-range-data.json.gz](#).

Validation loss by update — observed low-high range



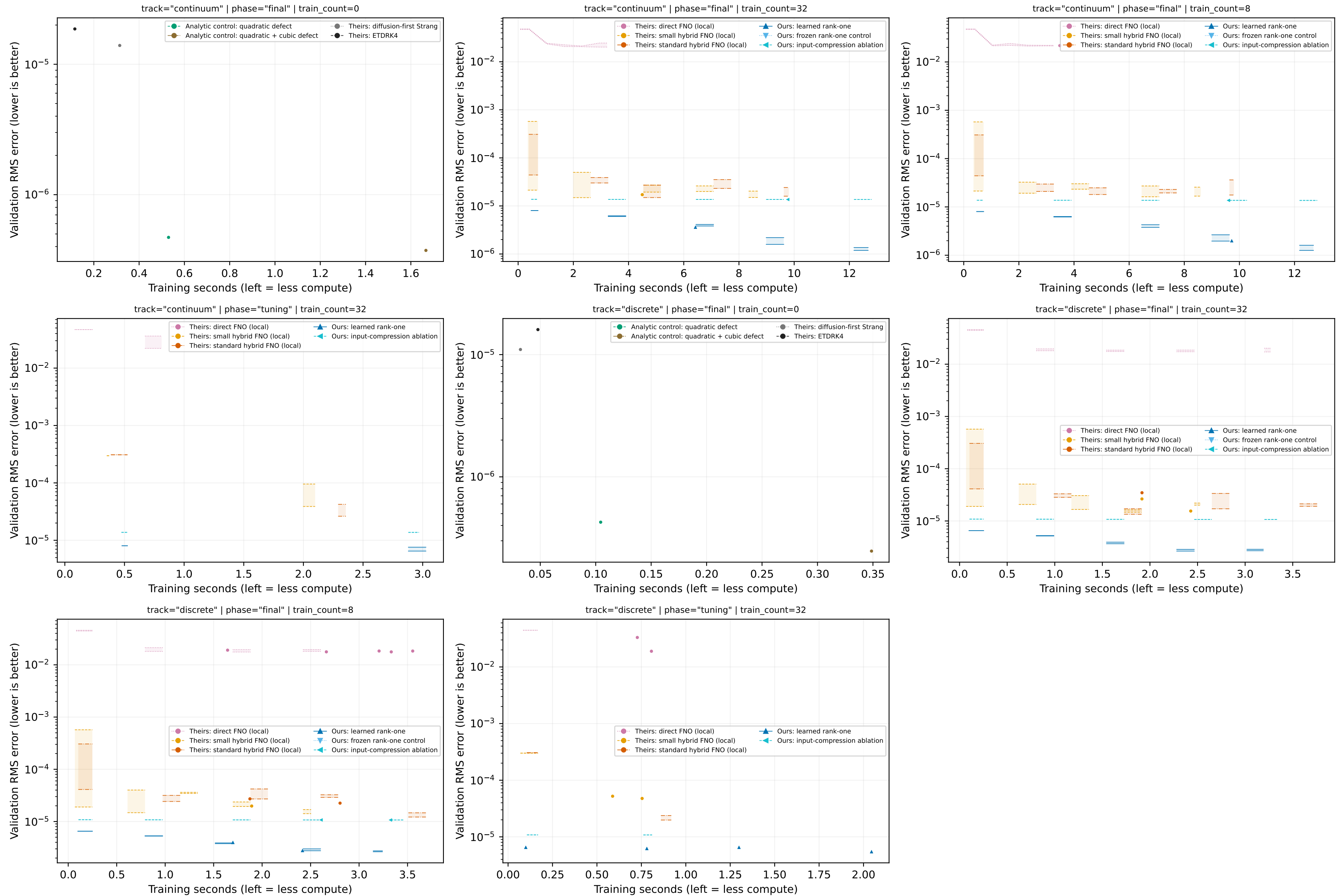
HOW TO READ: Y: lower validation loss is better on the same held-out objective. X: fewer updates to the same loss is preferable; more updates alone is not a win. Validation selection does not substitute for fresh confirmation.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Observed min/max in up to 20 shared x-windows across available seeds/trials; tuning includes all settings. Continuous connections interpolate window summaries, not additional observations or confidence intervals. Phase/data budgets stay separate; empty windows and single-x checkpoints break bands. Exact bounds/counts: learning-range-data.json.gz.

Validation error versus training time — observed low-high range



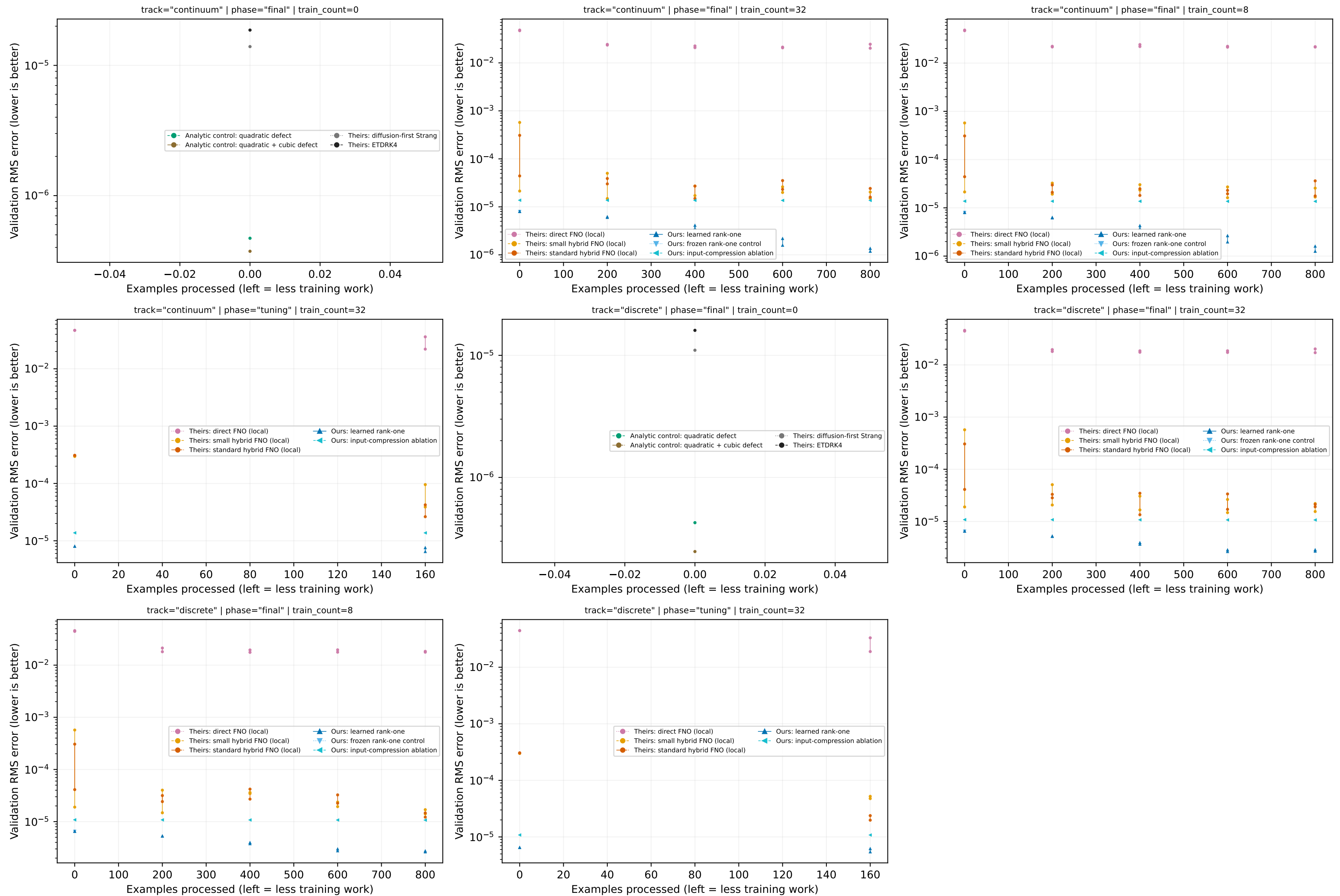
HOW TO READ: Lower-left is favorable: X less training time, Y lower validation RMS error. Compare the same target and validation cohort; a longer run may reach lower error. These are training costs, not inference latency.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Observed min/max in up to 20 shared x-windows across available seeds/trials; tuning includes all settings. Continuous connections interpolate window summaries, not additional observations or confidence intervals. Phase/data budgets stay separate; empty windows and single-x checkpoints break bands. Exact bounds/counts: learning-range-data.json.gz.

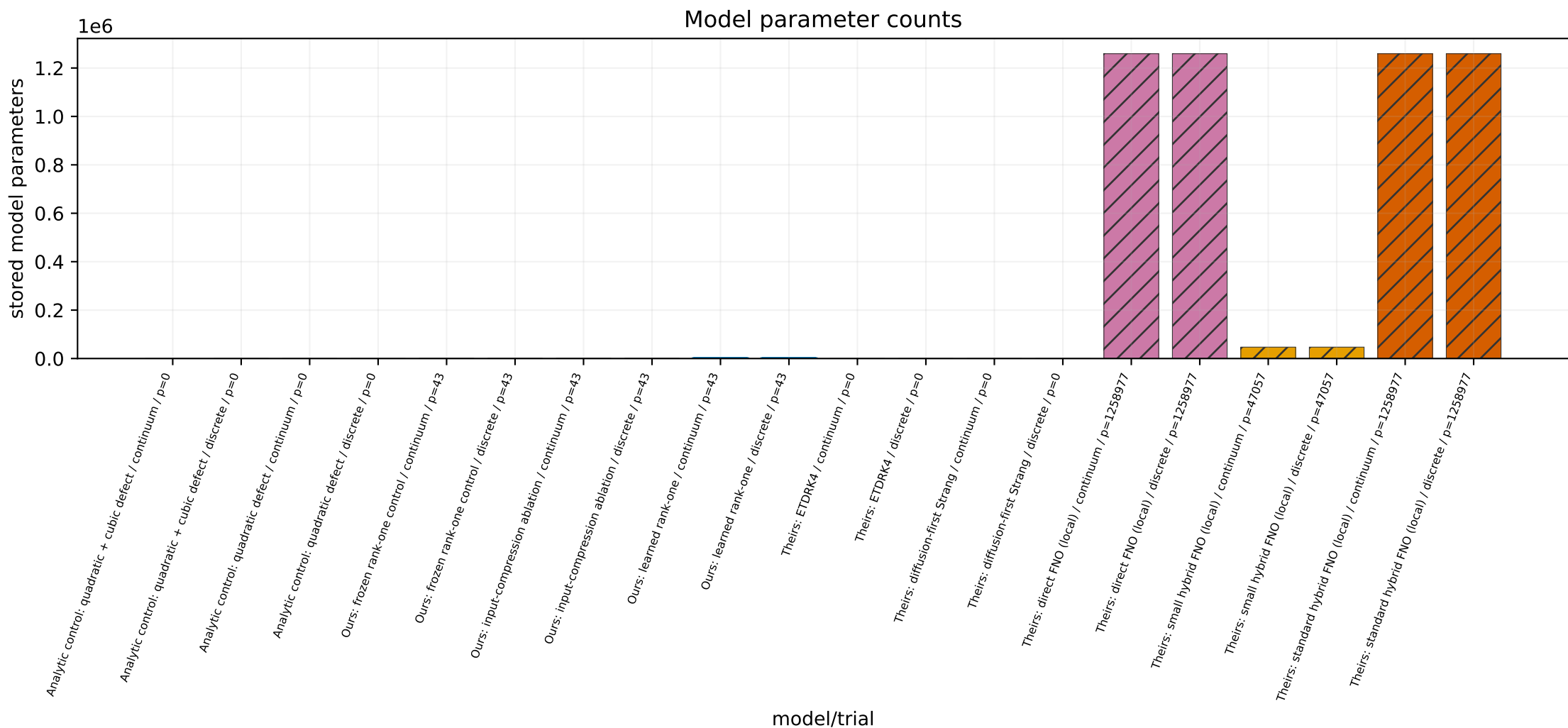
Validation error versus examples seen — observed low-high range



HOW TO READ: Lower-left is favorable: X fewer training-example presentations, Y lower validation RMS. Examples seen includes repeated exposure, not just independent fields. Compare like targets; this alone does not establish a sample-complexity law. METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Observed min/max in up to 20 shared x-windows across available seeds/trials; tuning includes all settings. Continuous connections interpolate window summaries, not additional observations or confidence intervals. Phase/data budgets stay separate; empty windows and single-x checkpoints break bands. Exact bounds/counts: learning-range-data.json.gz.



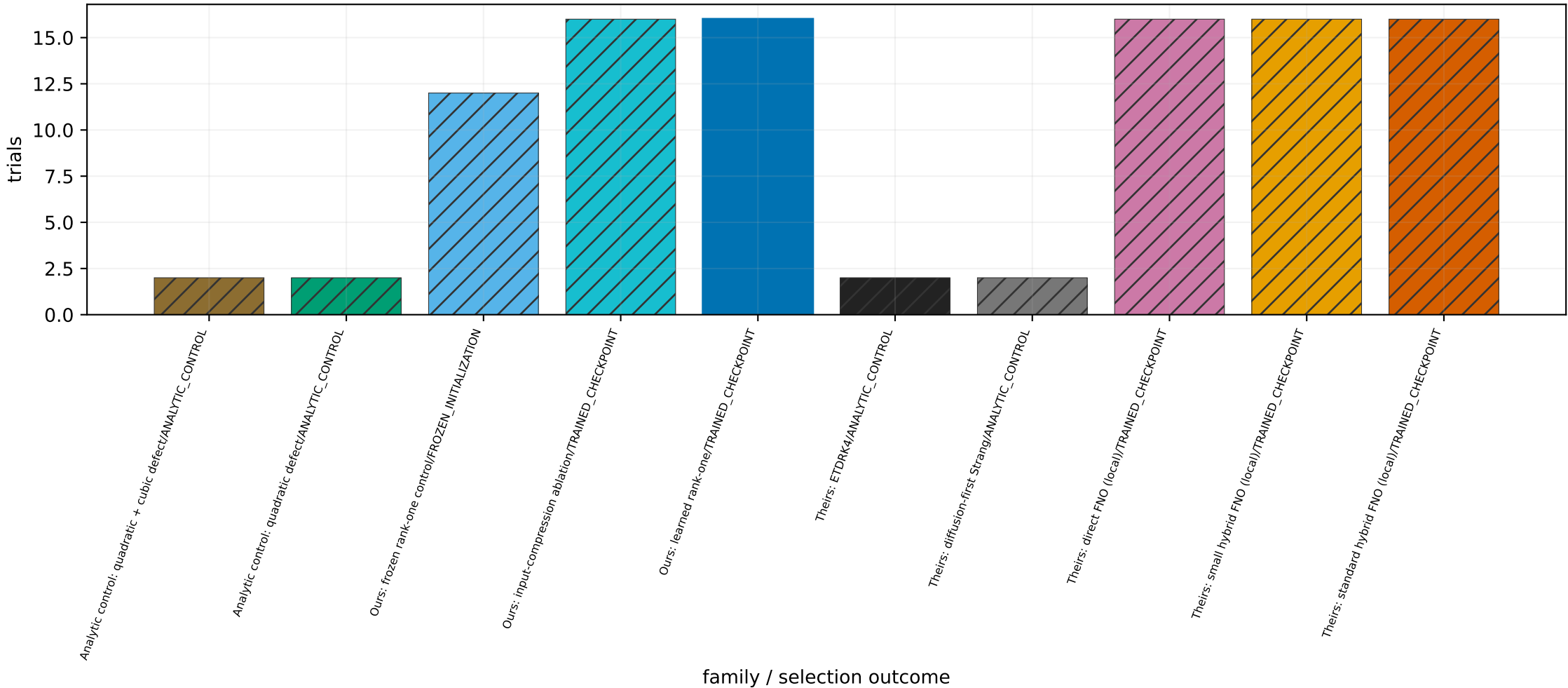
HOW TO READ: Y: lower counts mean a smaller stored model, not necessarily faster or more accurate. Capacity and accuracy can trade off; parameter count is not FLOPs or measured runtime. X lists methods/trials and has no favorable direction.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Identical parameter counts collapse only for display by family/track; every model ID is retained in chart data. Parameter count is not FLOPs or runtime.

Baseline adequacy and checkpoint selections



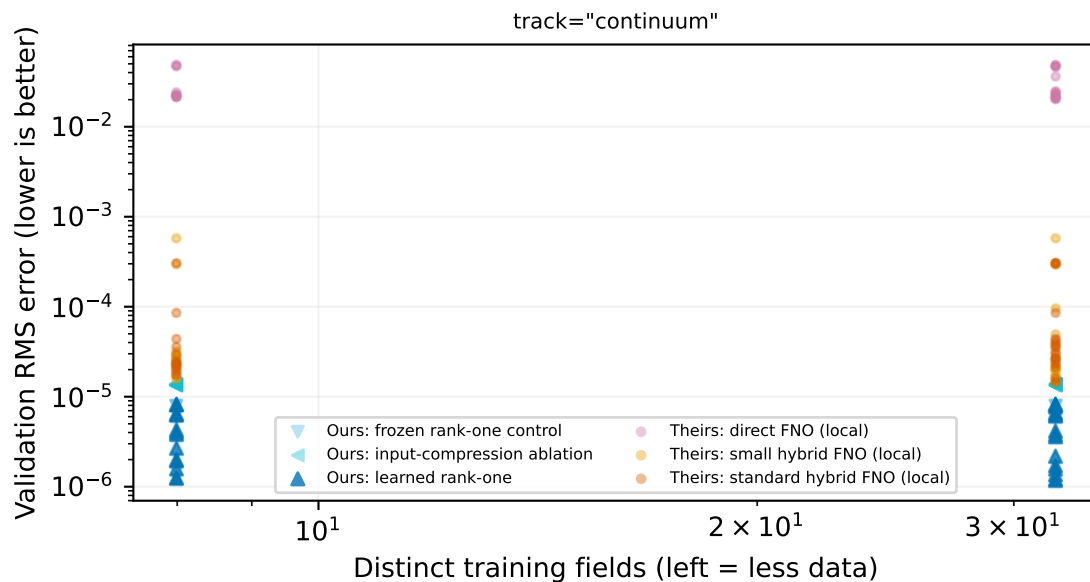
HOW TO READ: Y counts selection outcomes, not method quality. A selected trained checkpoint can show improvement over its initialization, but does not establish convergence. An initialization selection is not evidence of successful learning.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

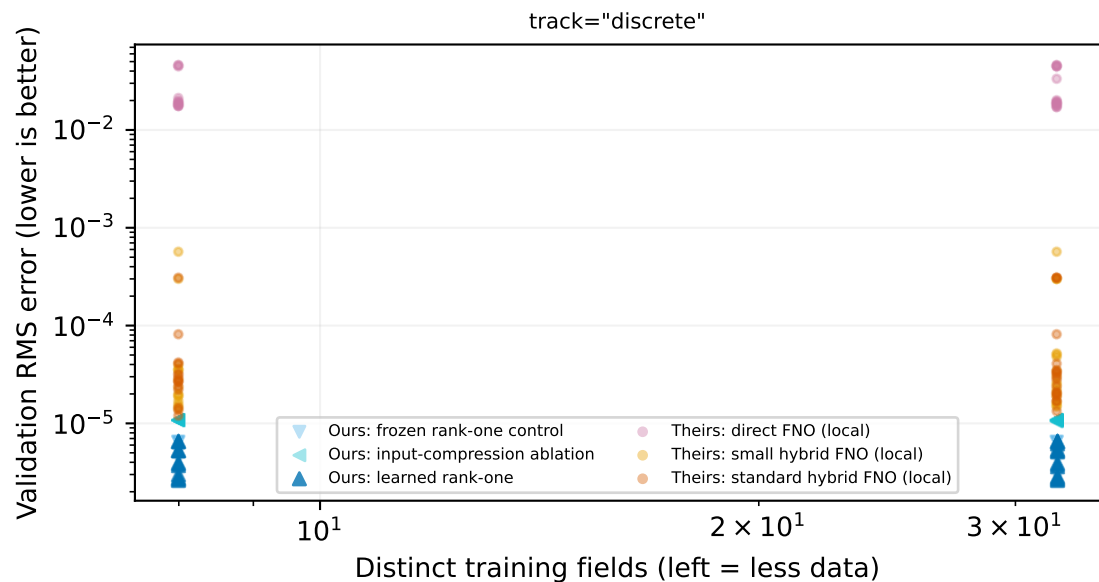
MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Initialization selection is not evidence of learned improvement. A weak trained control does not establish superiority over a published method.

Data efficiency validation observations



All 176 finite points; unavailable/nonpositive 6274



All 176 finite points; unavailable/nonpositive 6274

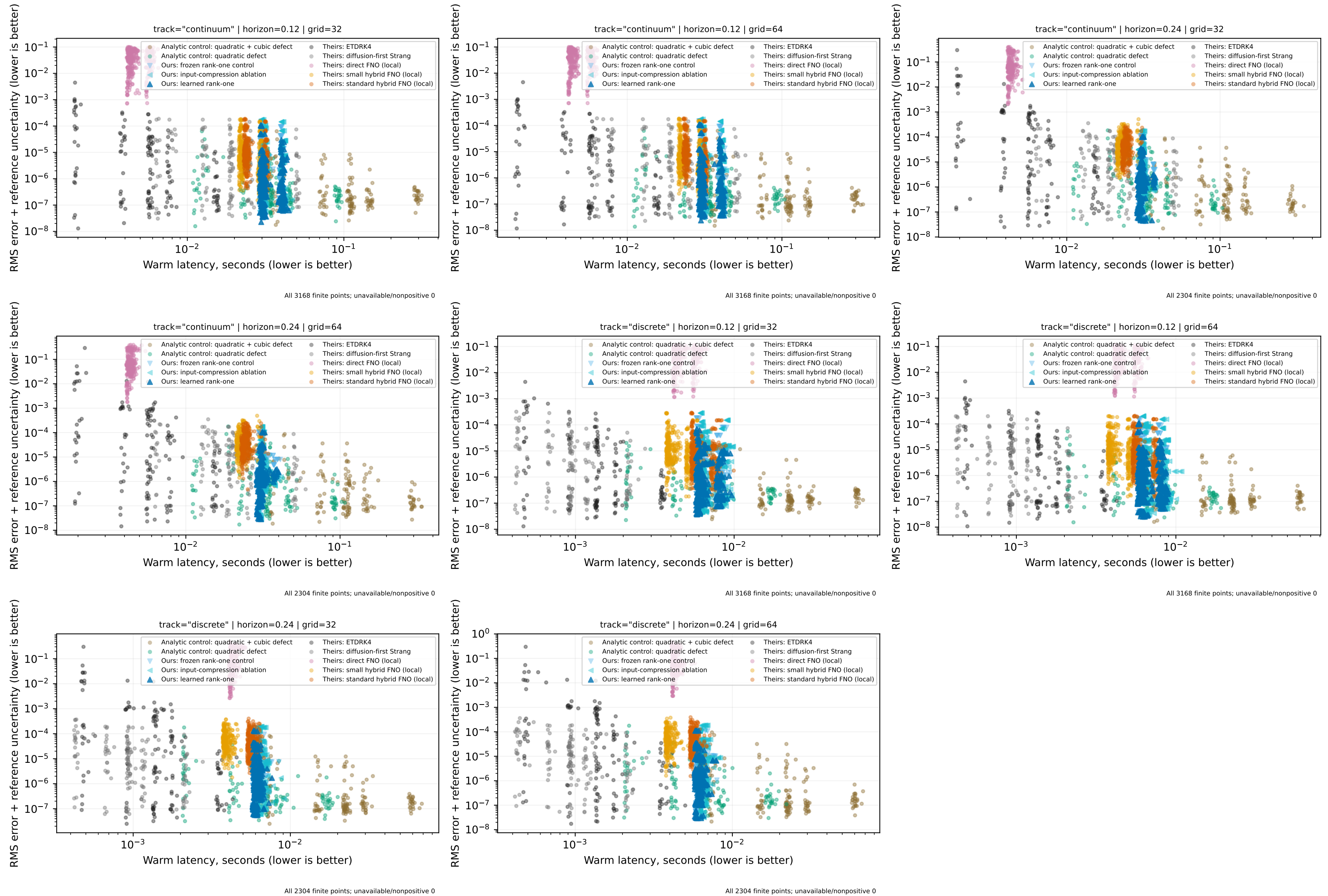
HOW TO READ: Lower-left is favorable: X fewer training fields, Y lower validation RMS error. Compare like targets and training budgets; updates from one run are repeated observations, not additional independent fields.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: All observed updates shown; this is descriptive, not an independent sample count. Training and confirmation remain disjoint.

Work precision rms error with reference uncertainty



HOW TO READ: Lower-left is favorable: X less warmed runtime, Y lower RMS error plus reference uncertainty. Both axes are logarithmic; compare within the same track, grid and final time. Speed is useful only when the required accuracy is met.

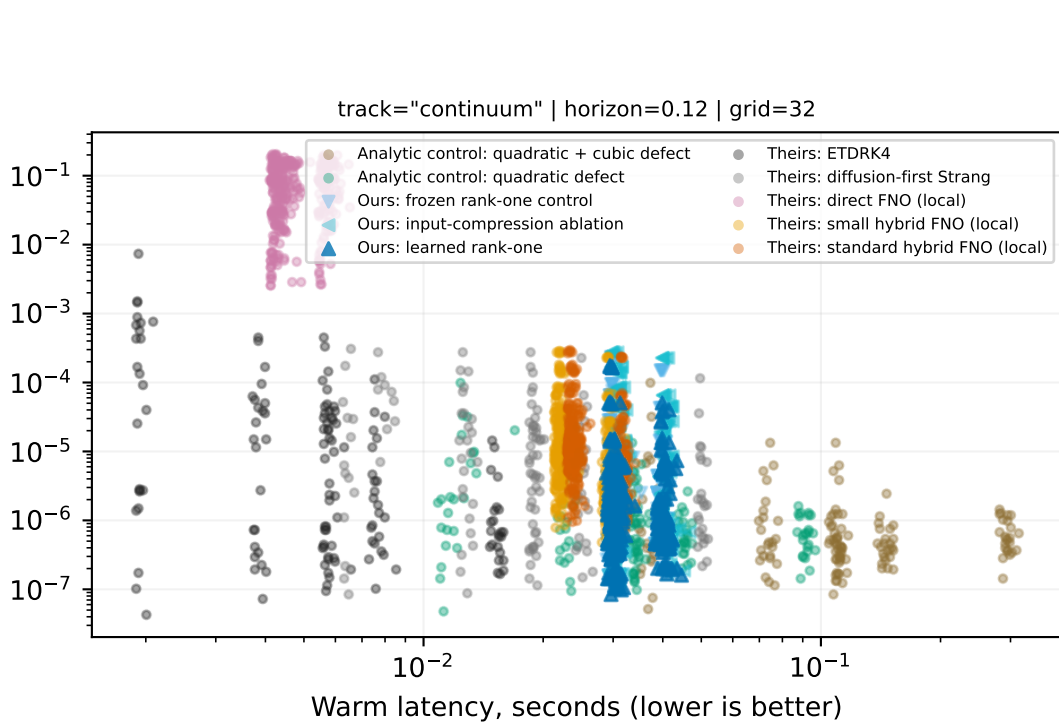
METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: All endpoint schedules retained; each panel has one spatial target and final time. Reference-informed frontier selection is post hoc, not a deployable policy.

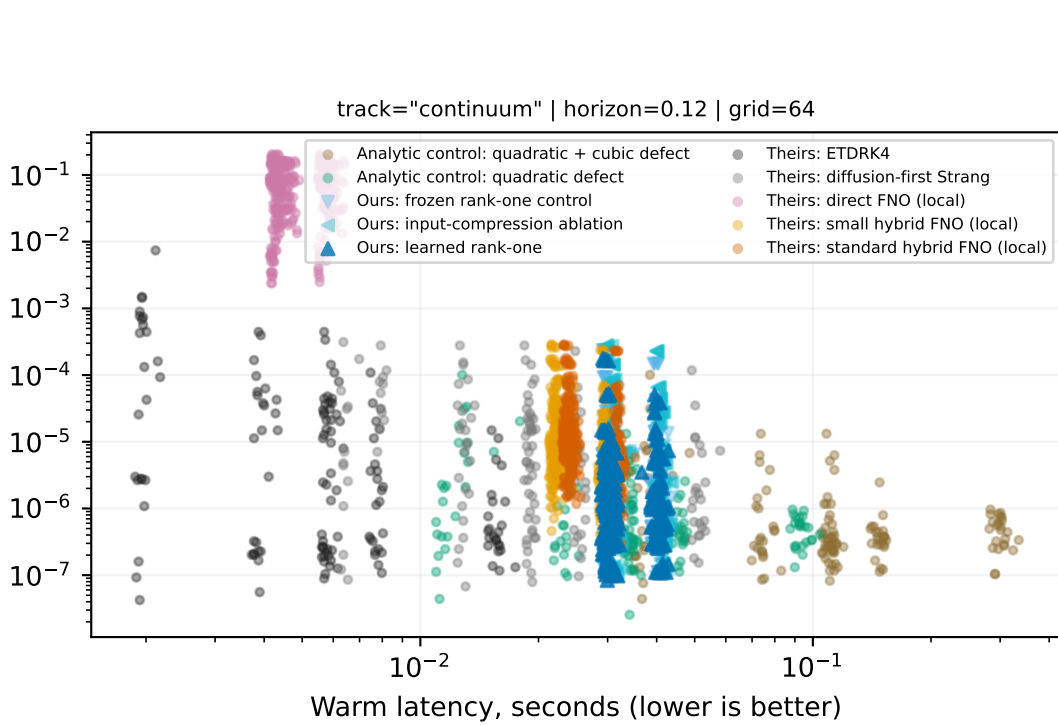
Work precision max error with reference uncertainty

Maximum error + reference uncertainty (lower is better)Maximum error + reference uncertainty (lower is better)



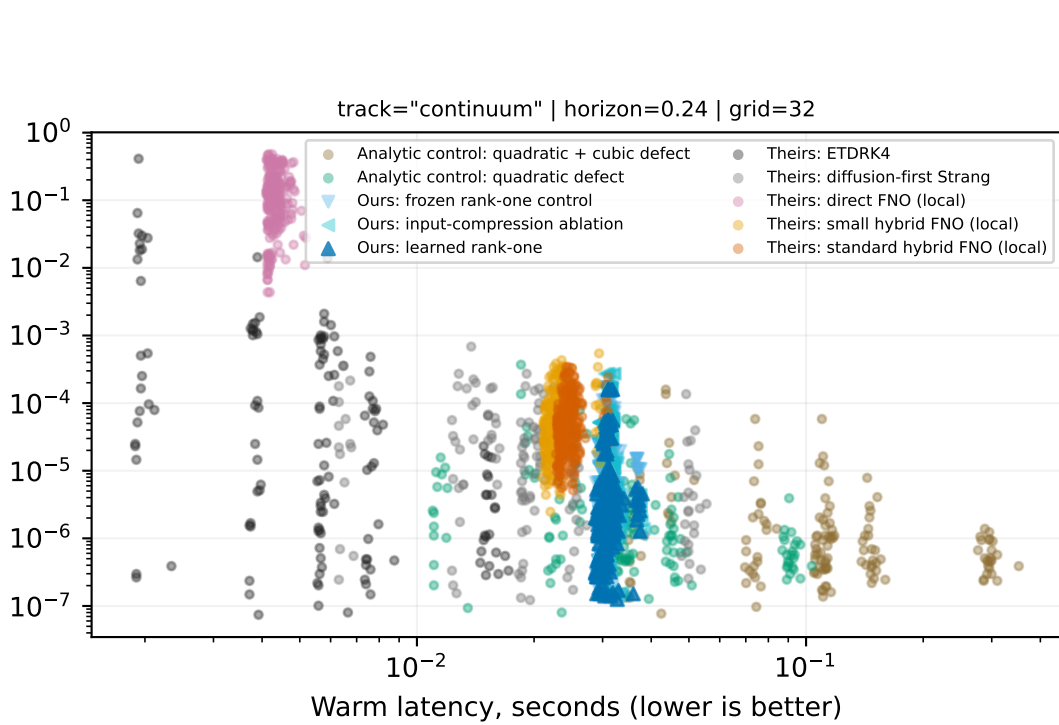
All 3168 finite points; unavailable/nonpositive 0

Maximum error + reference uncertainty (lower is better)Maximum error + reference uncertainty (lower is better)

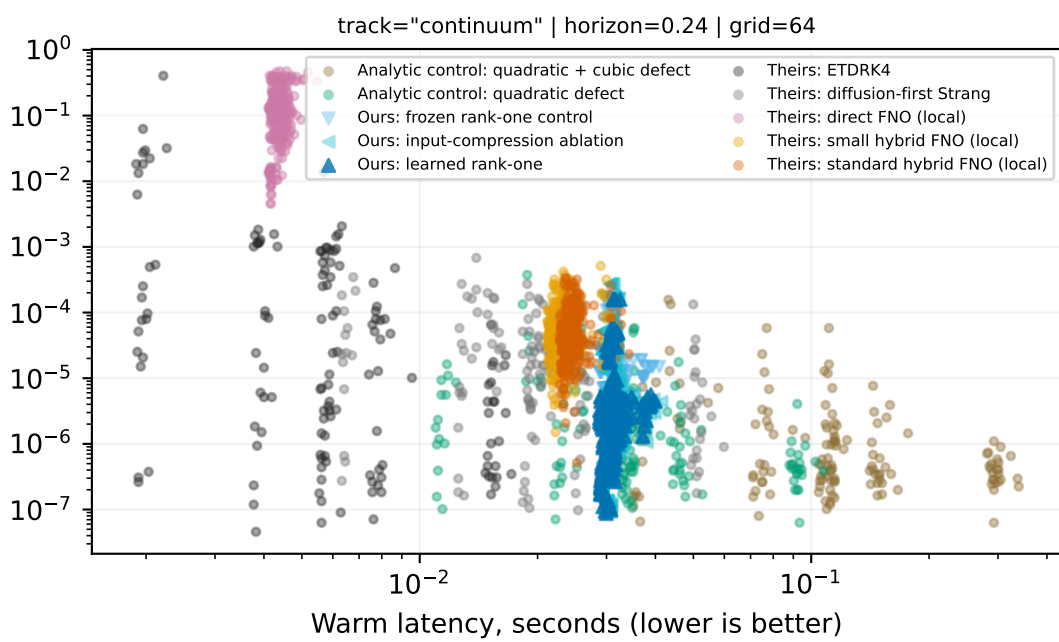


All 3168 finite points; unavailable/nonpositive 0

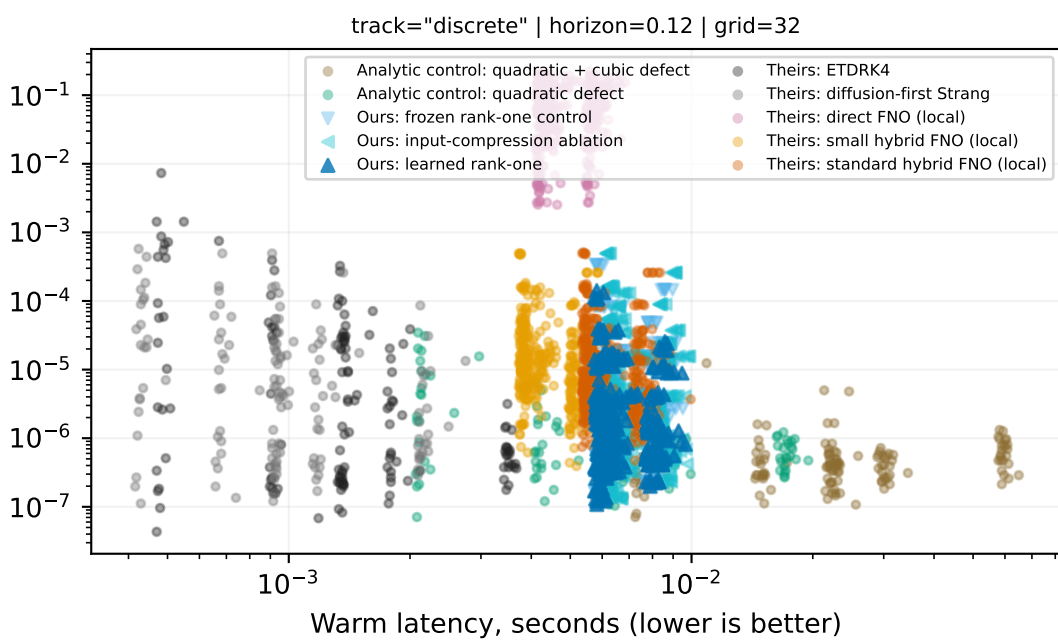
Maximum error + reference uncertainty (lower is better)Maximum error + reference uncertainty (lower is better)



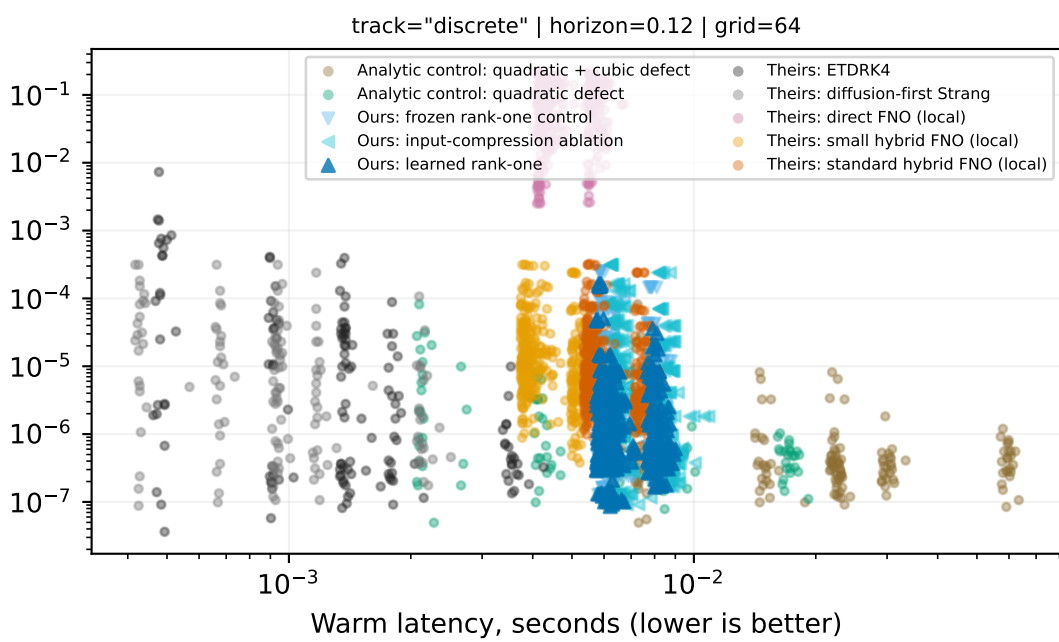
All 2304 finite points; unavailable/nonpositive 0



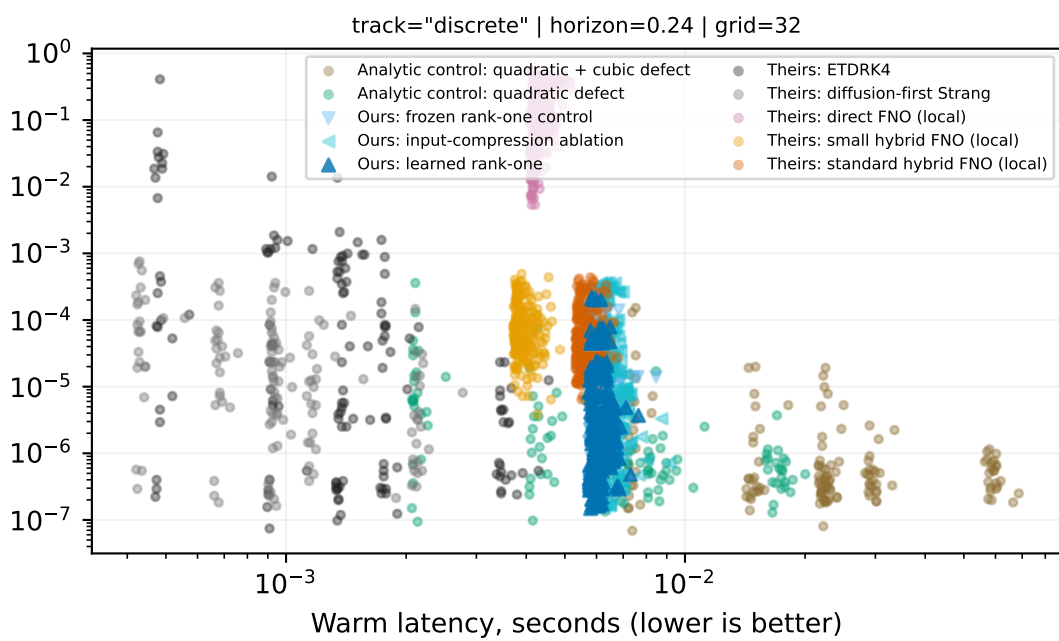
All 2304 finite points; unavailable/nonpositive 0



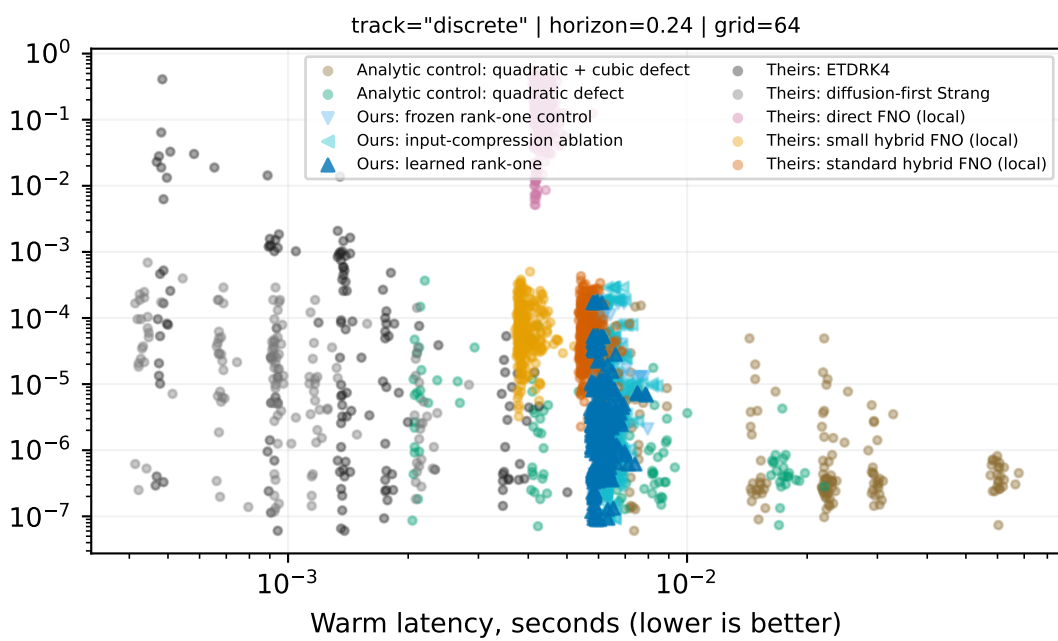
All 3168 finite points; unavailable/nonpositive 0



All 3168 finite points; unavailable/nonpositive 0



All 2304 finite points; unavailable/nonpositive 0



All 2304 finite points; unavailable/nonpositive 0

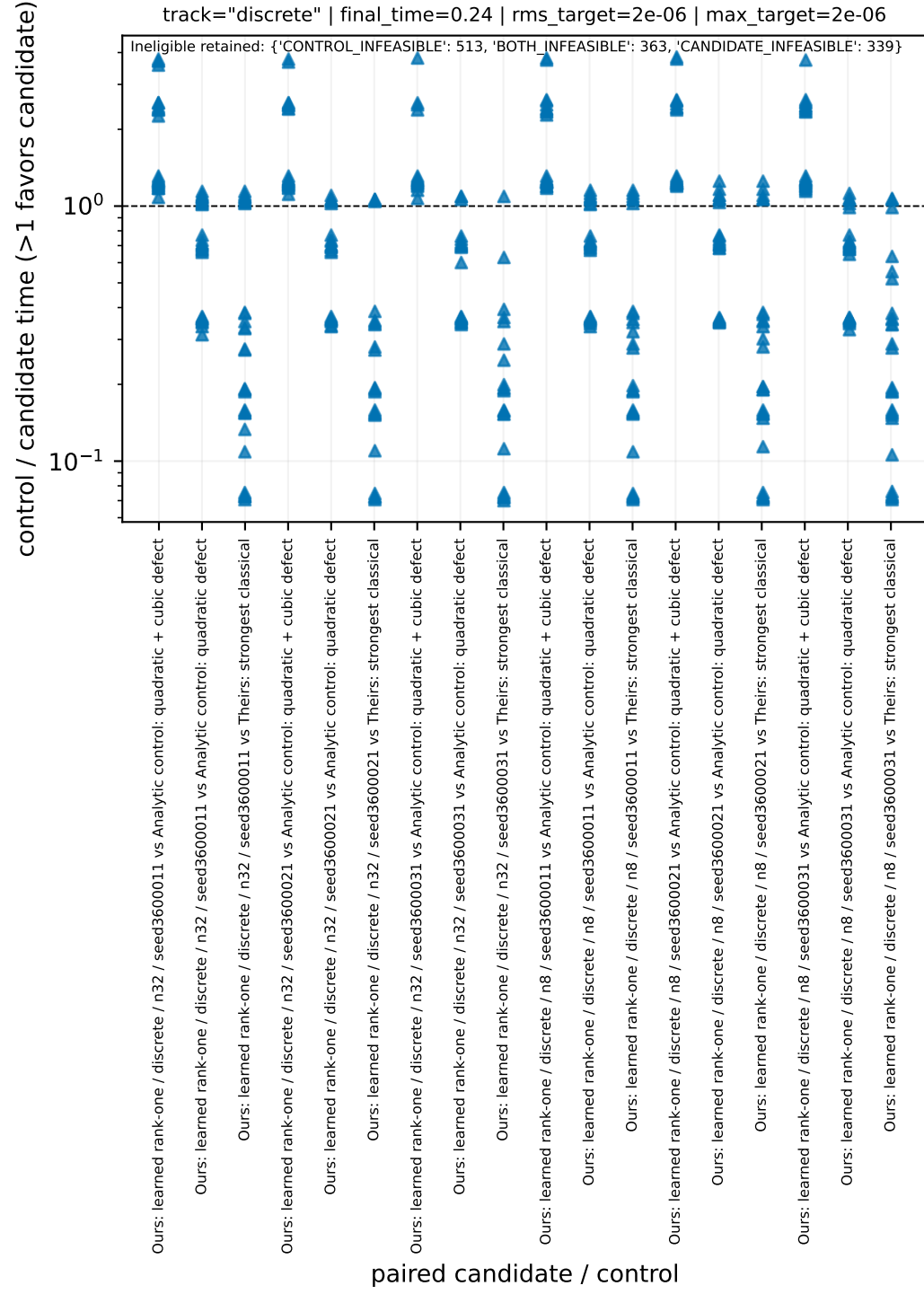
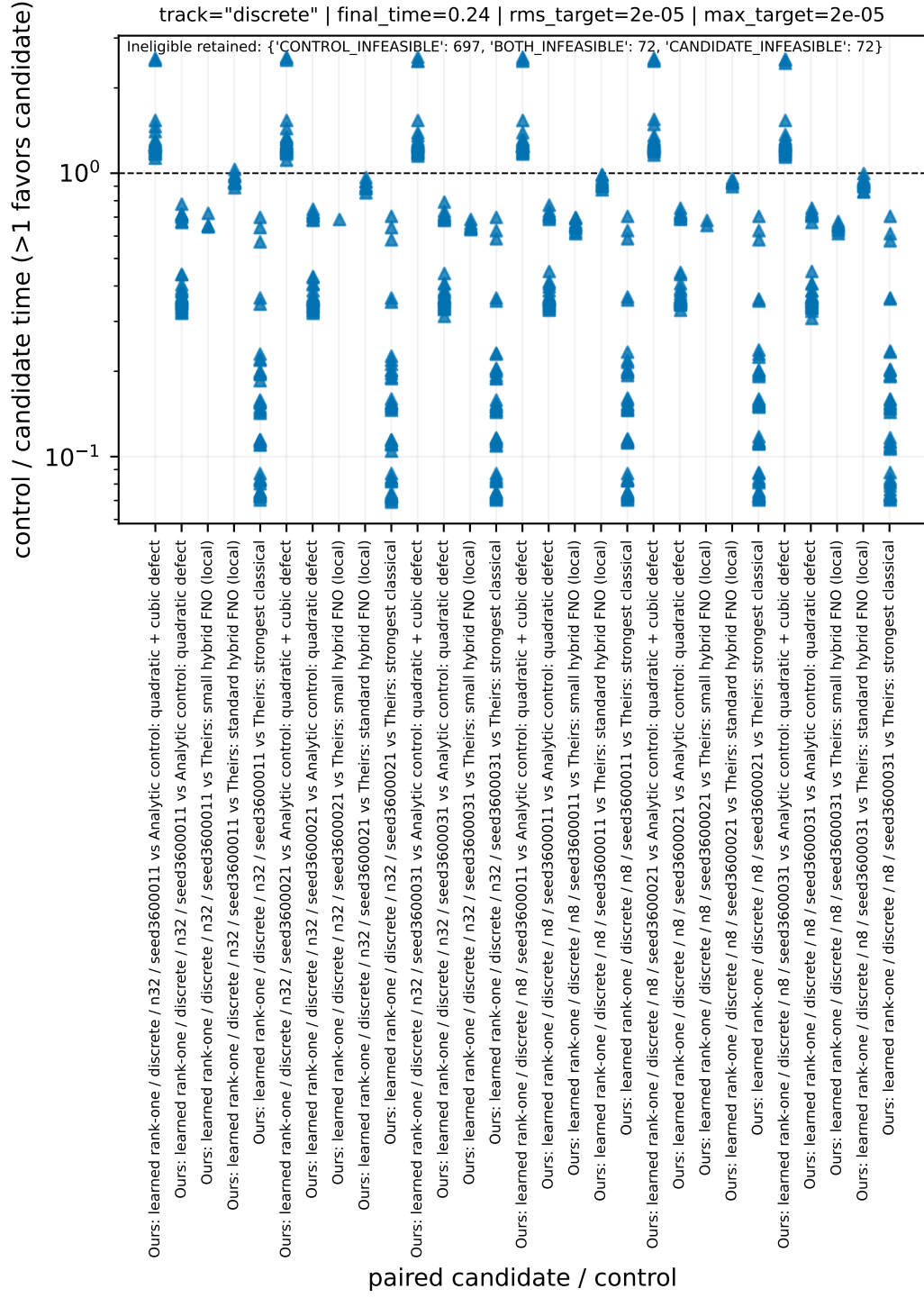
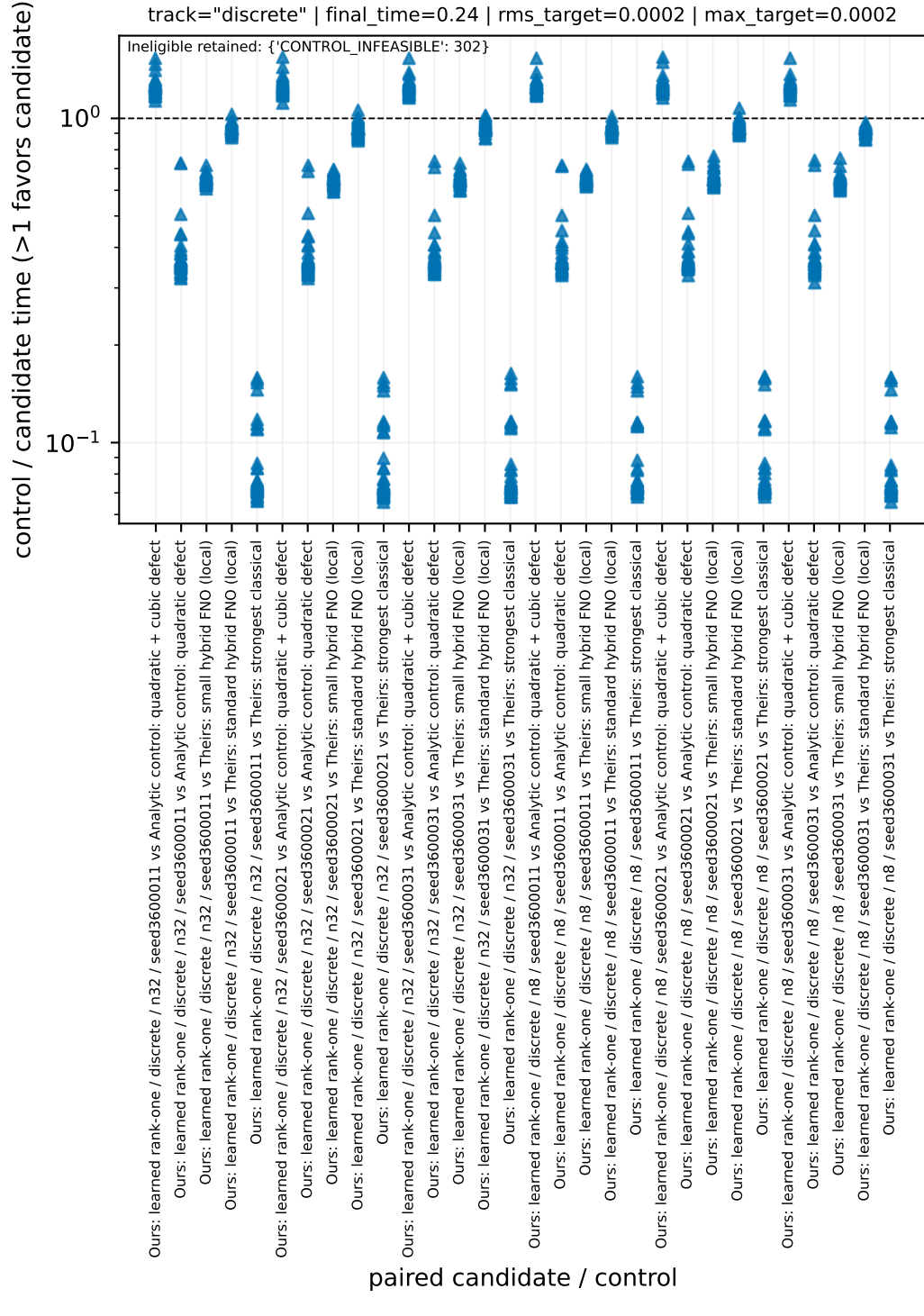
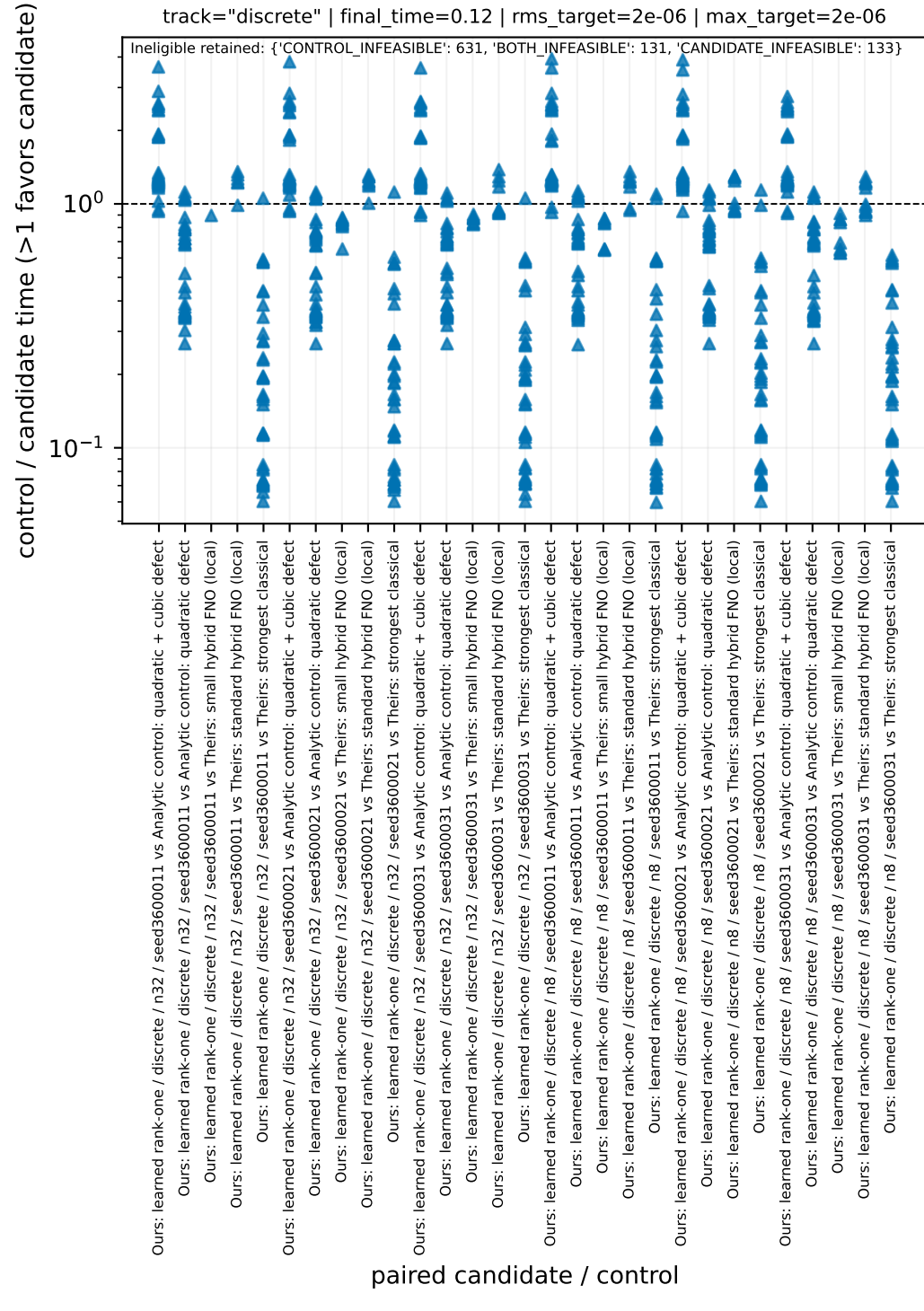
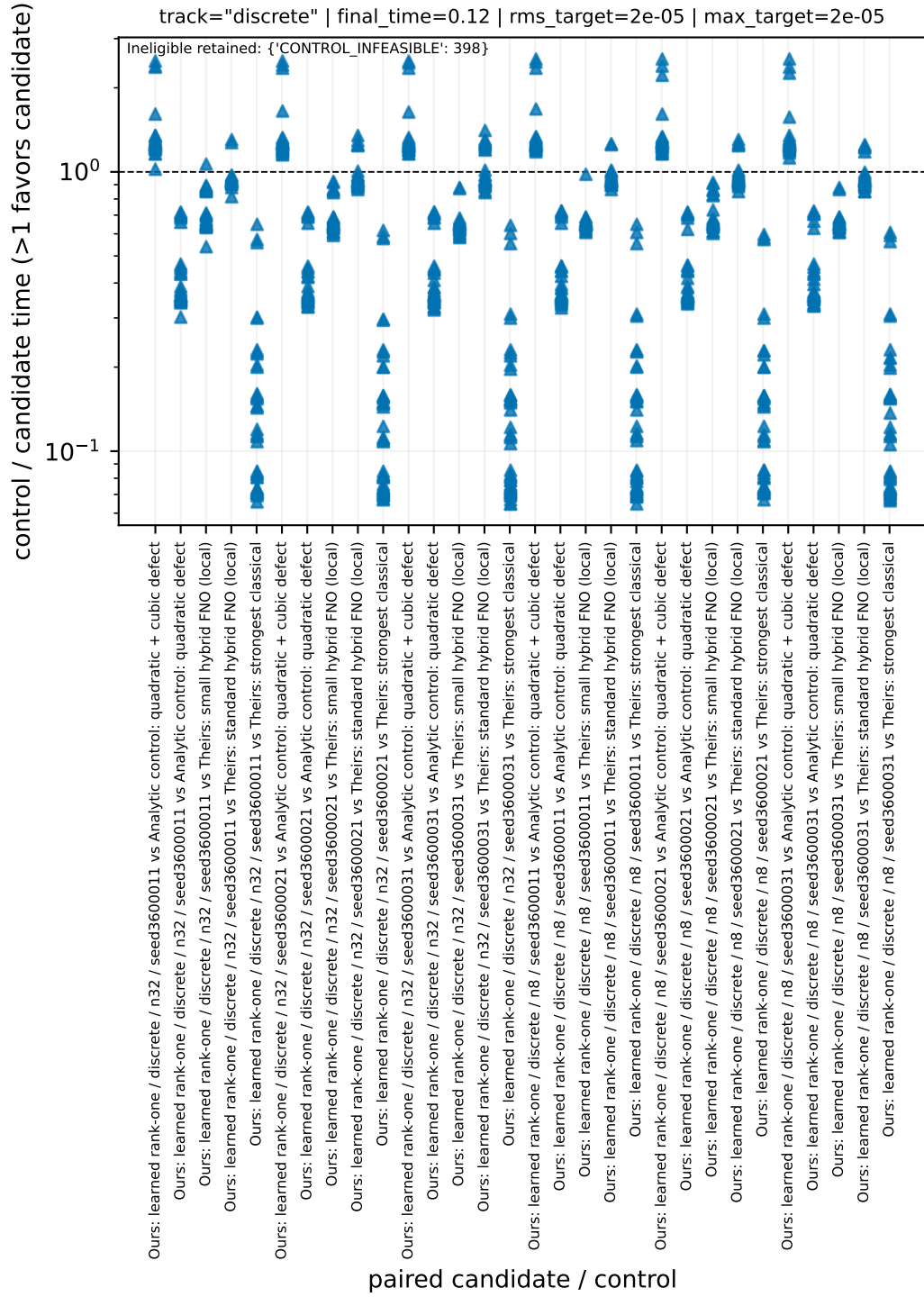
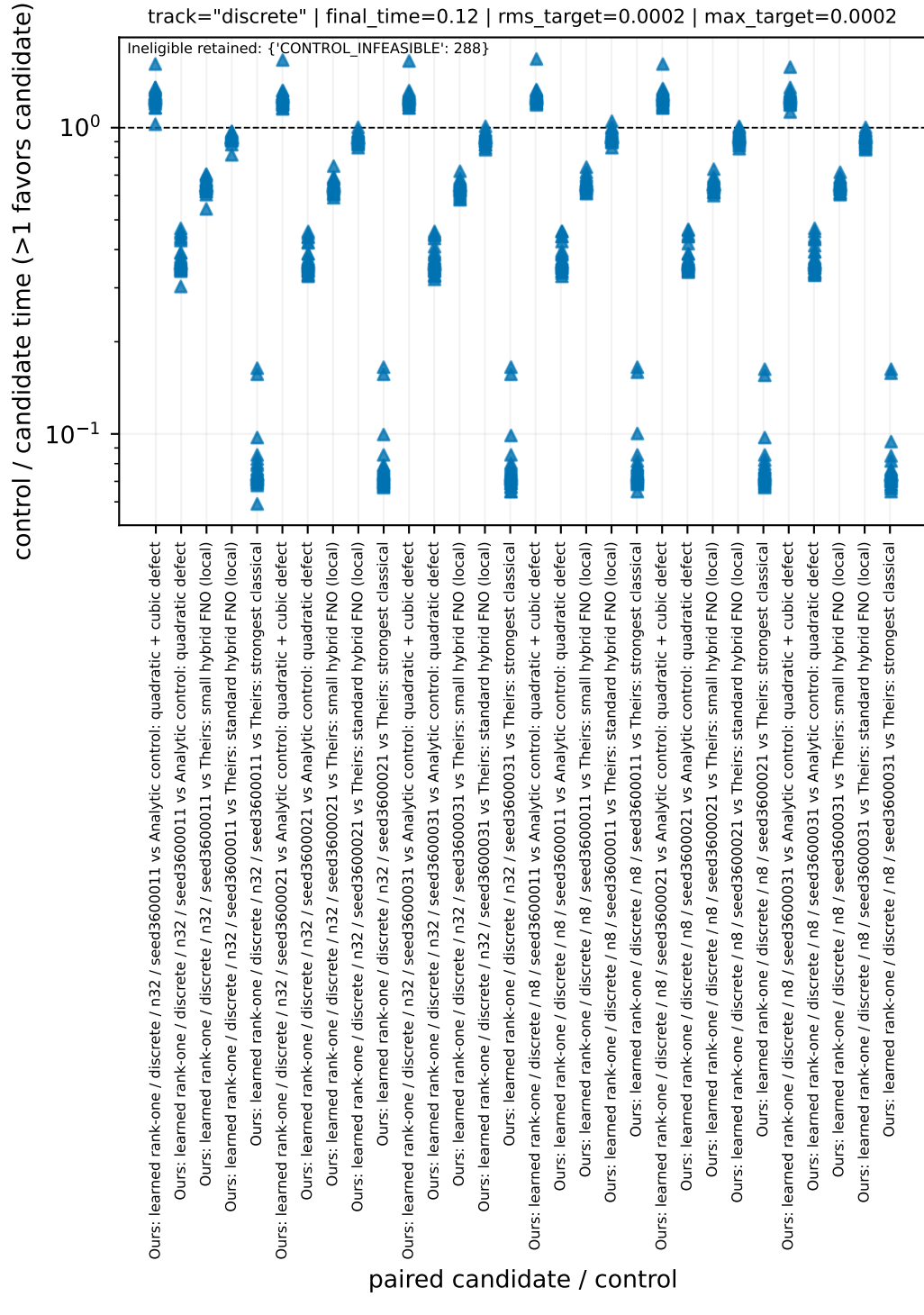
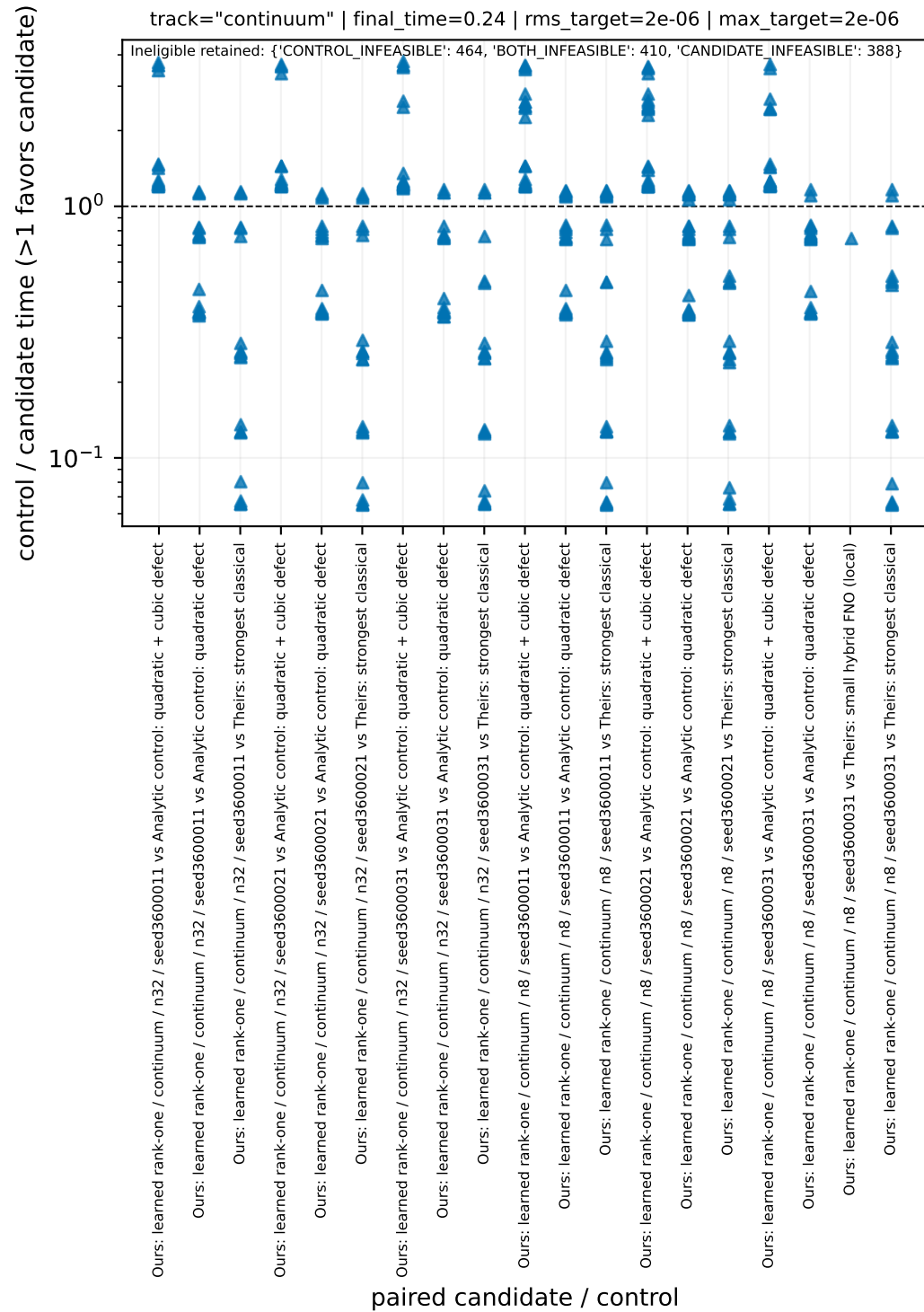
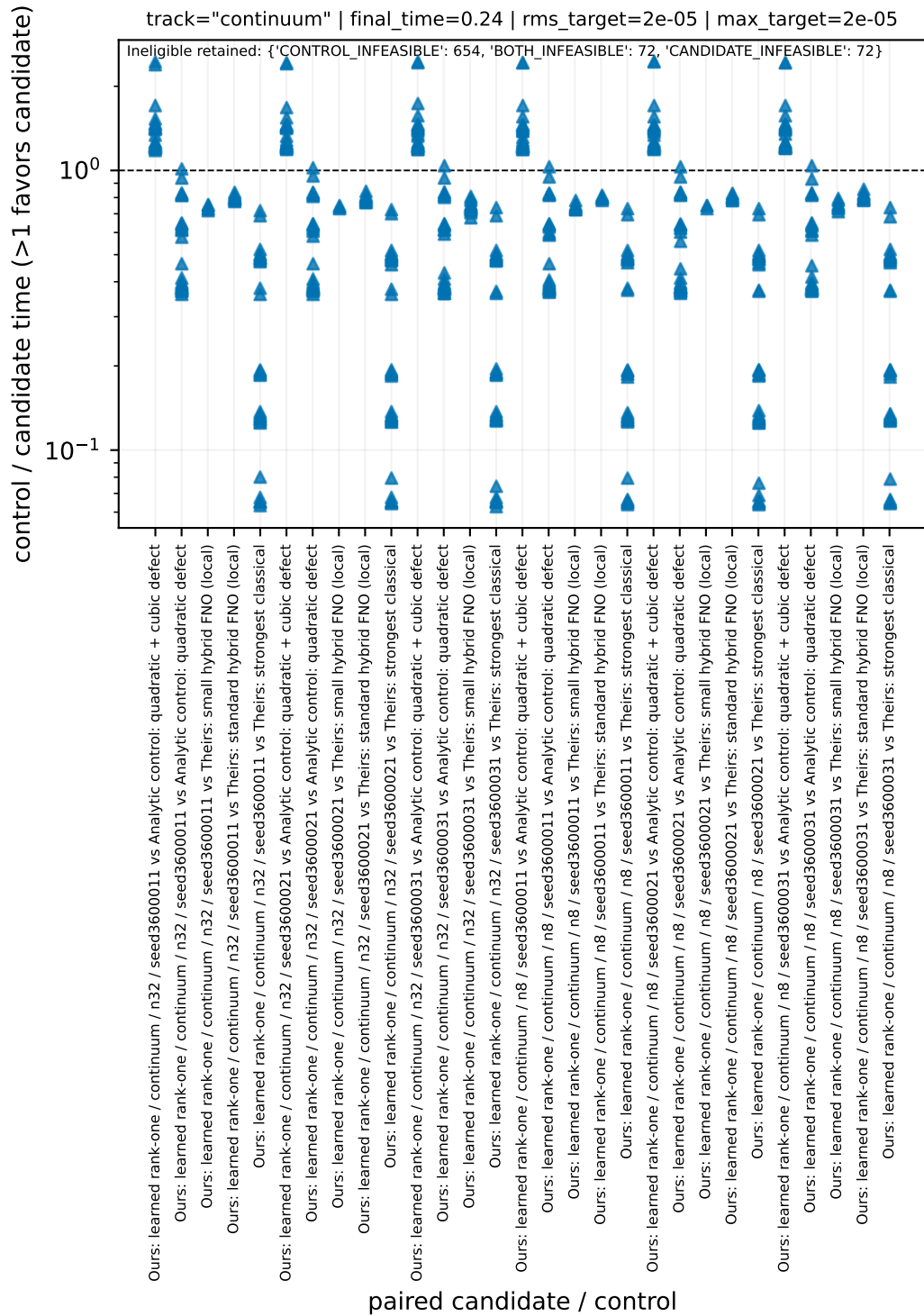
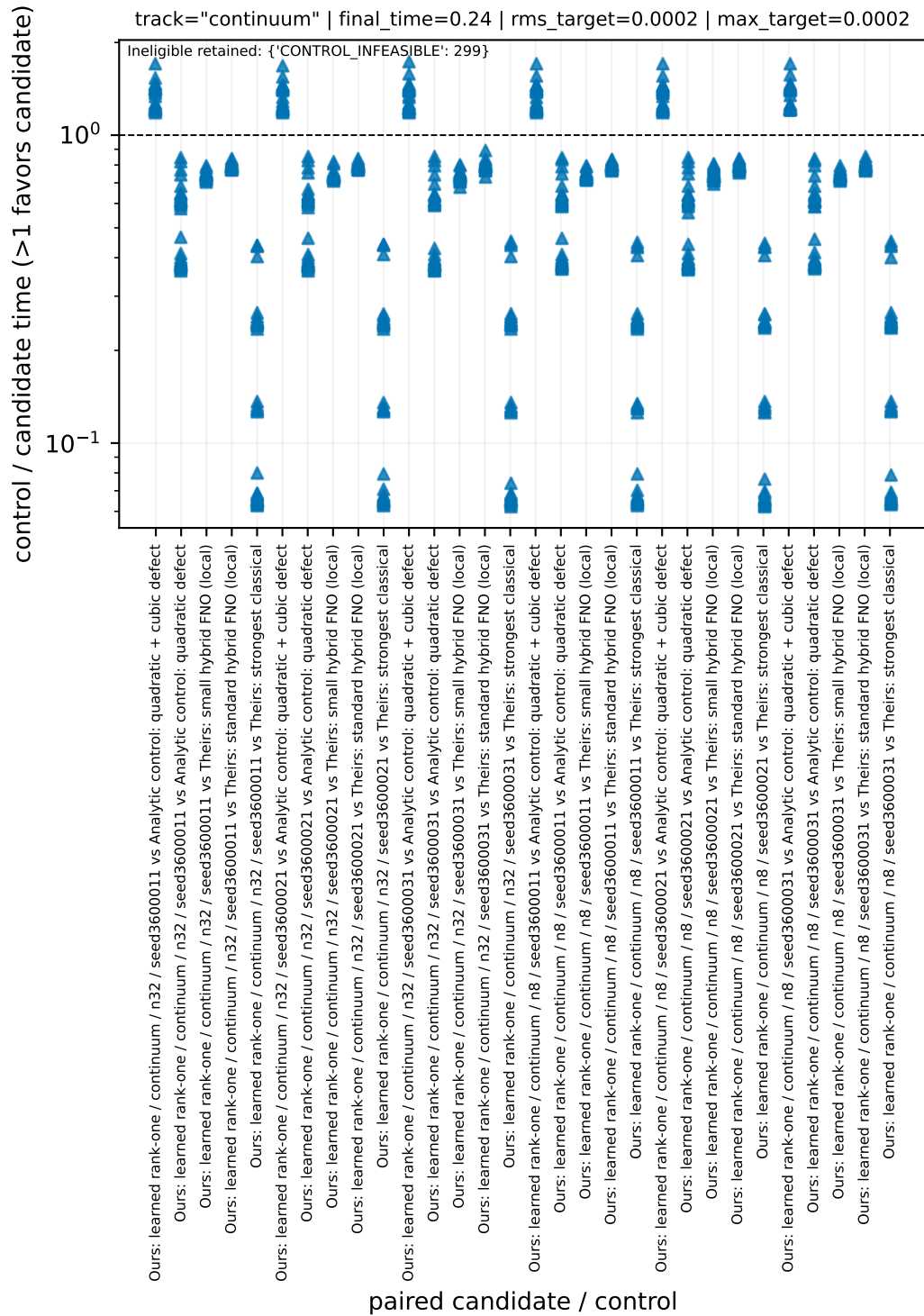
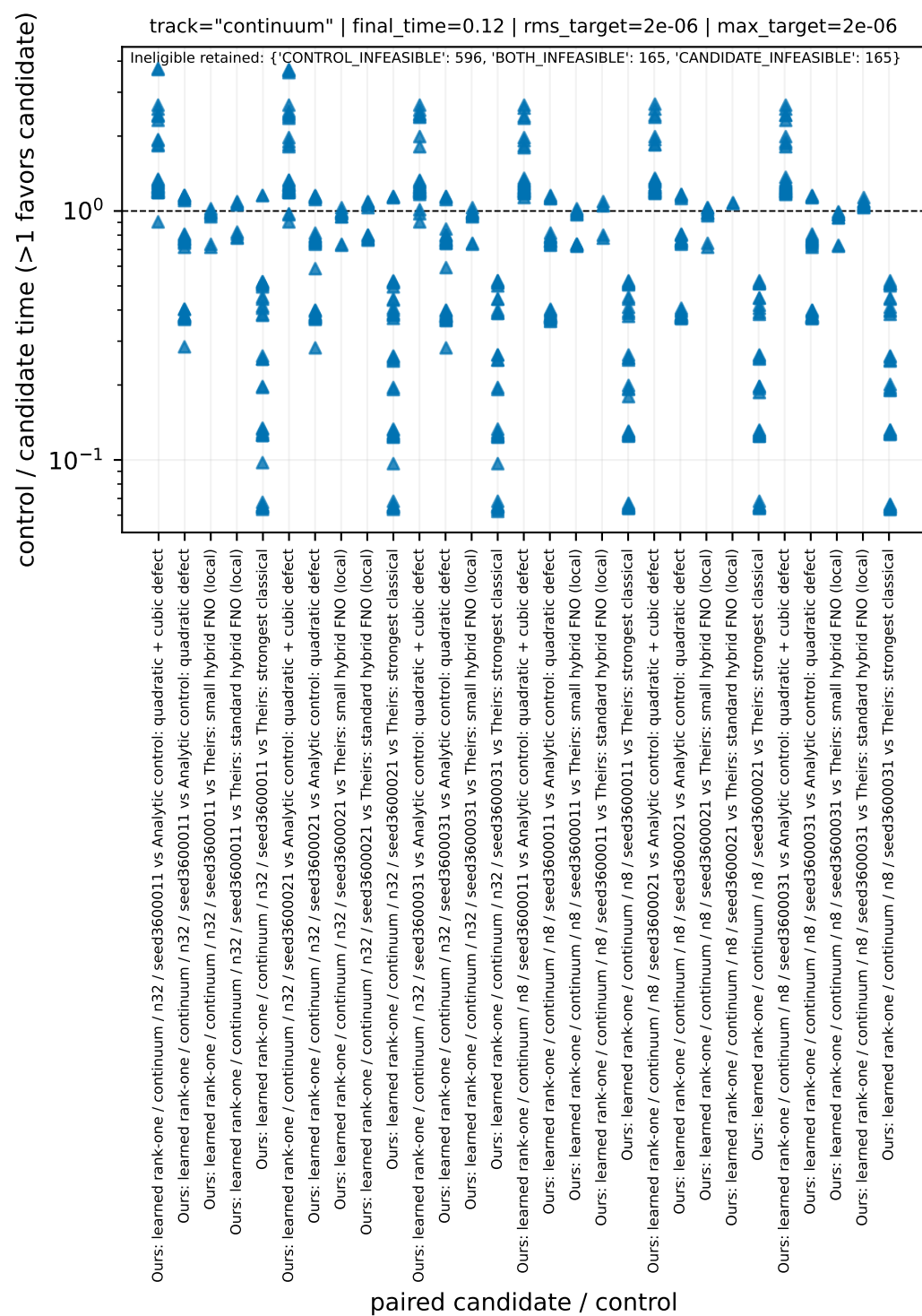
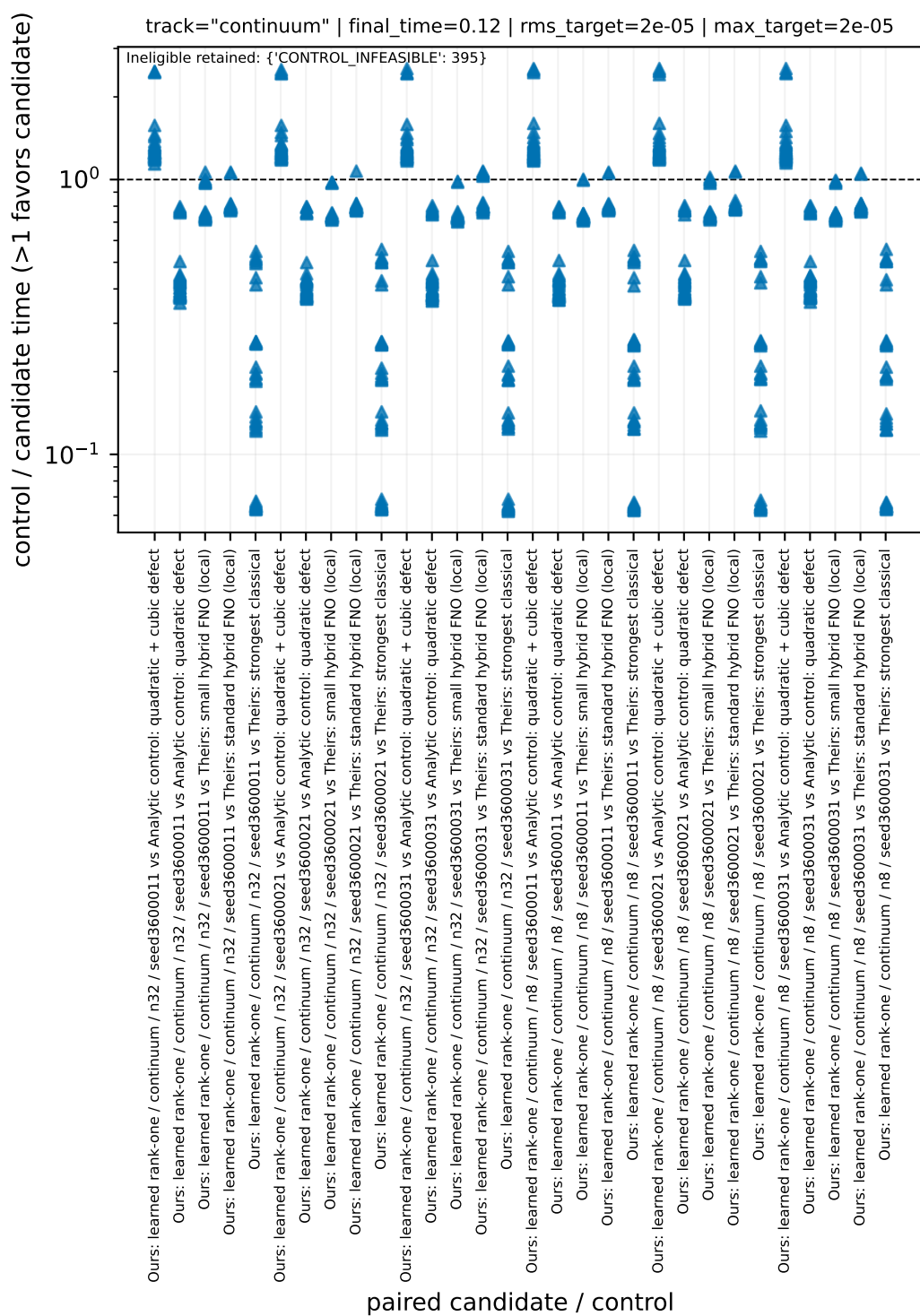
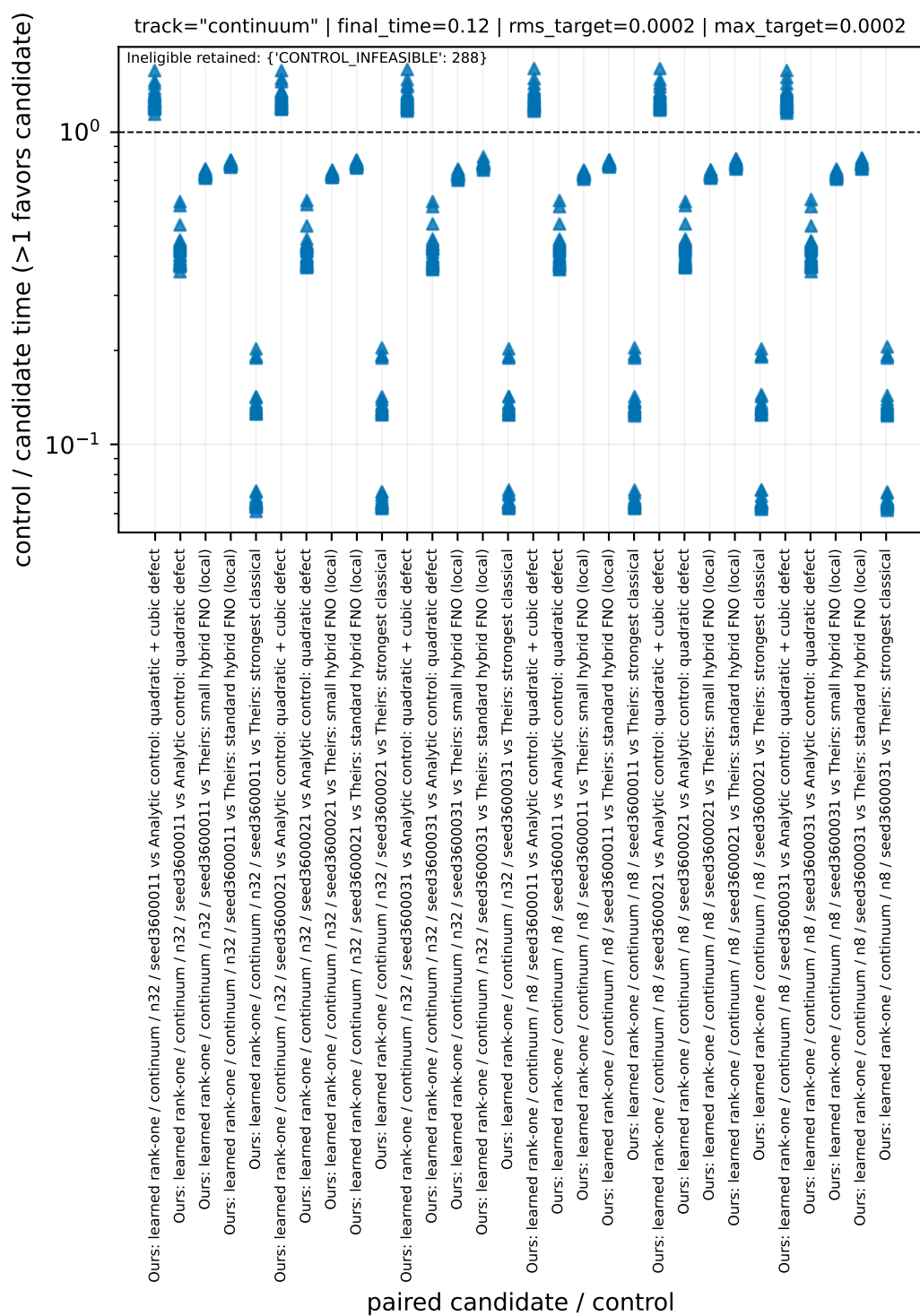
HOW TO READ: Lower-left is favorable: X less warmed runtime, Y lower maximum error plus reference uncertainty. Both axes are logarithmic; compare within the same track, grid and final time. Meeting RMS alone does not establish the maximum-error target.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: All endpoint schedules retained; each panel has one spatial target and final time. Reference-informed frontier selection is post hoc, not a deployable policy.

Matched target paired speed distributions



HOW TO READ: Y = control time / candidate time: above 1 favors the candidate, below 1 favors the control, and 1 is a tie. Higher is better for the named candidate. The axis is logarithmic; ineligible cases are not speed wins.

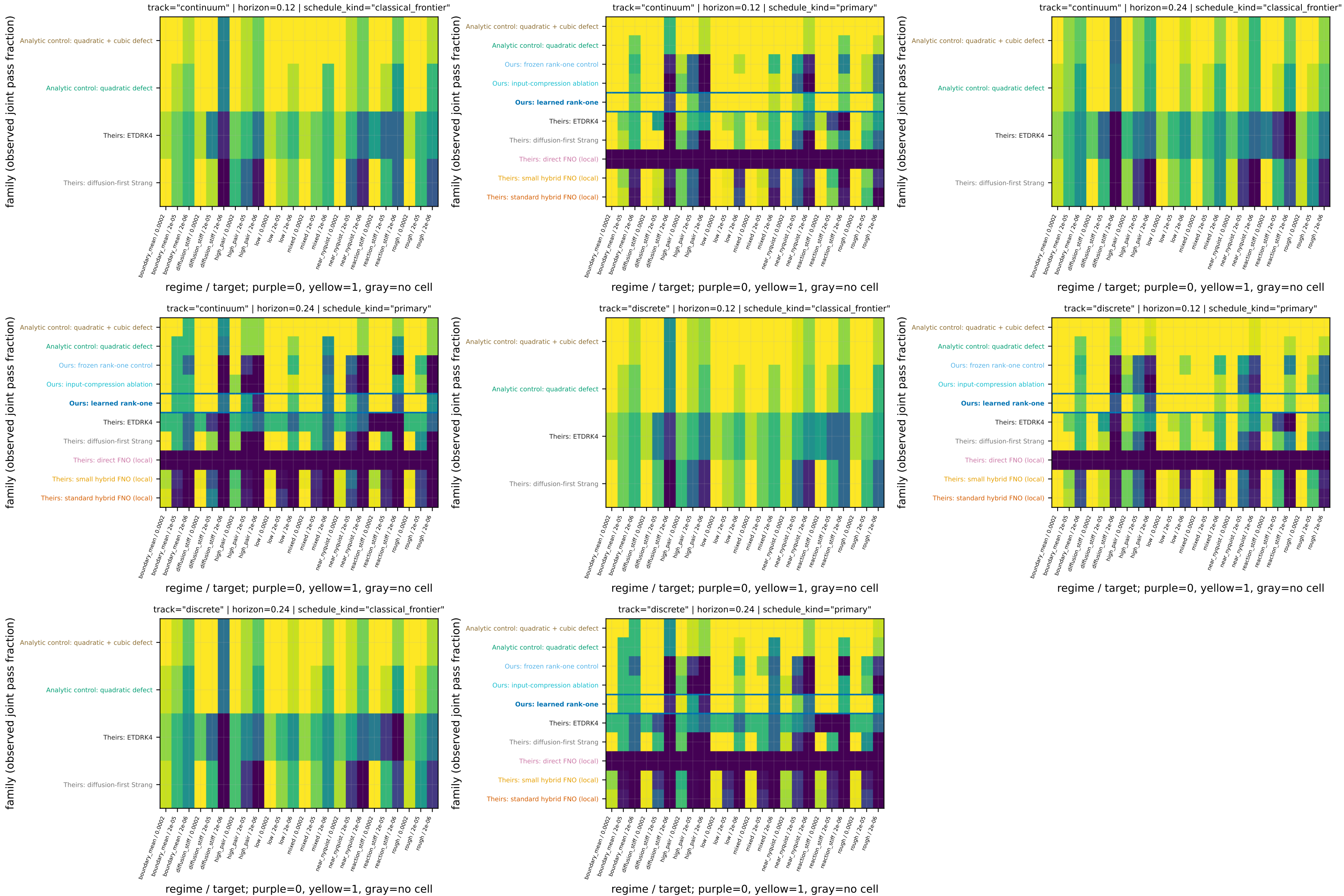
METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate

styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Different final times and tolerances are never pooled. Repeated schedules/grids/seeds are paired evidence; use field-clustered inference in comparisons.json.

Stress regime and tolerance coverage



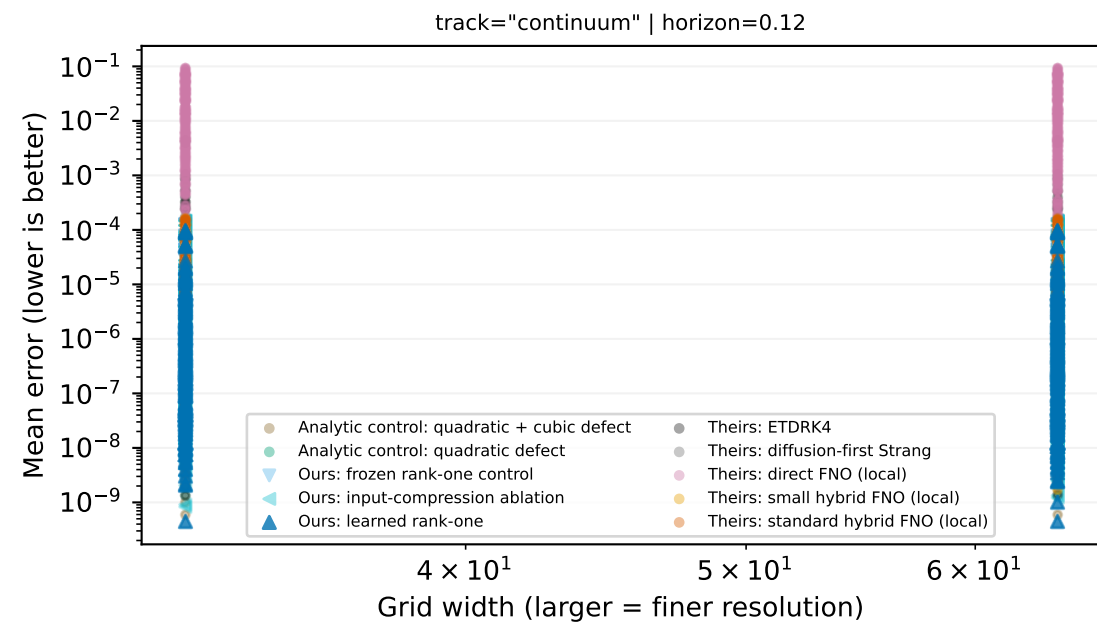
HOW TO READ: Color: higher joint pass fraction is better; yellow = 1, purple = 0. Gray is missing evidence, not a favorable result. X/Y identify regimes/targets and methods; their positions are not quality rankings.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

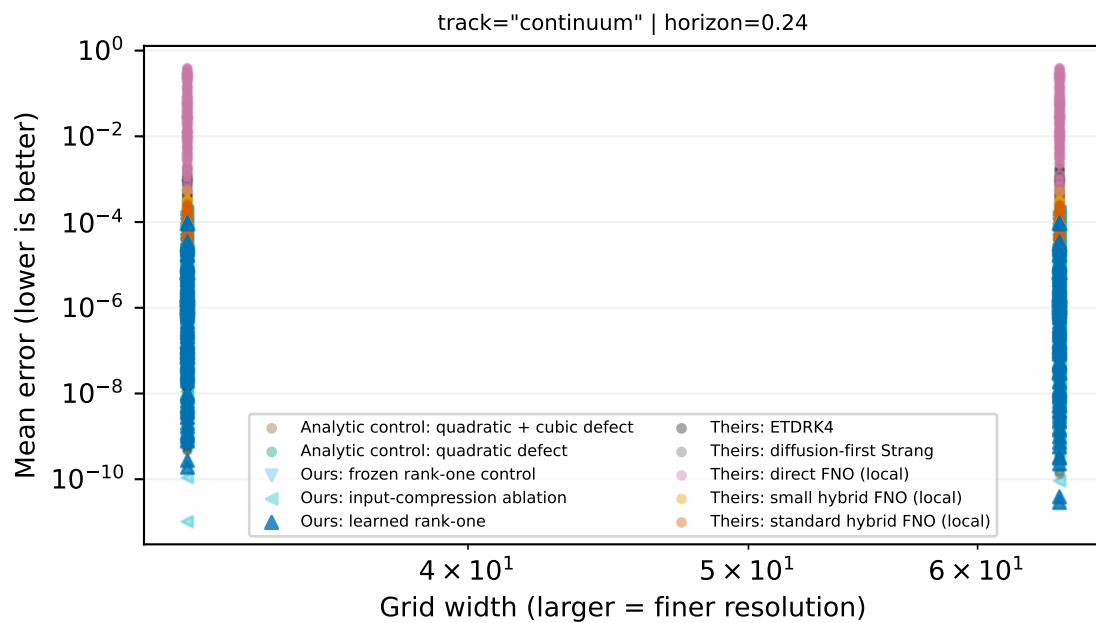
MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Normal, best and worst cases use frozen field regimes, not retrospective exclusion. Cells with no target evidence remain NA.

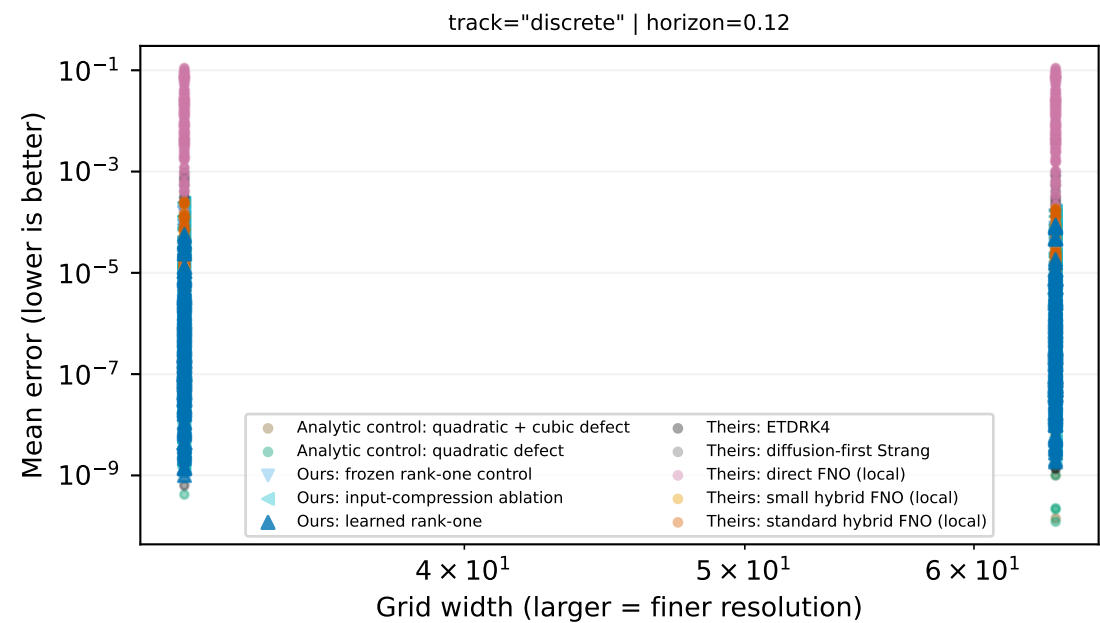
Mean error across spatial resolution



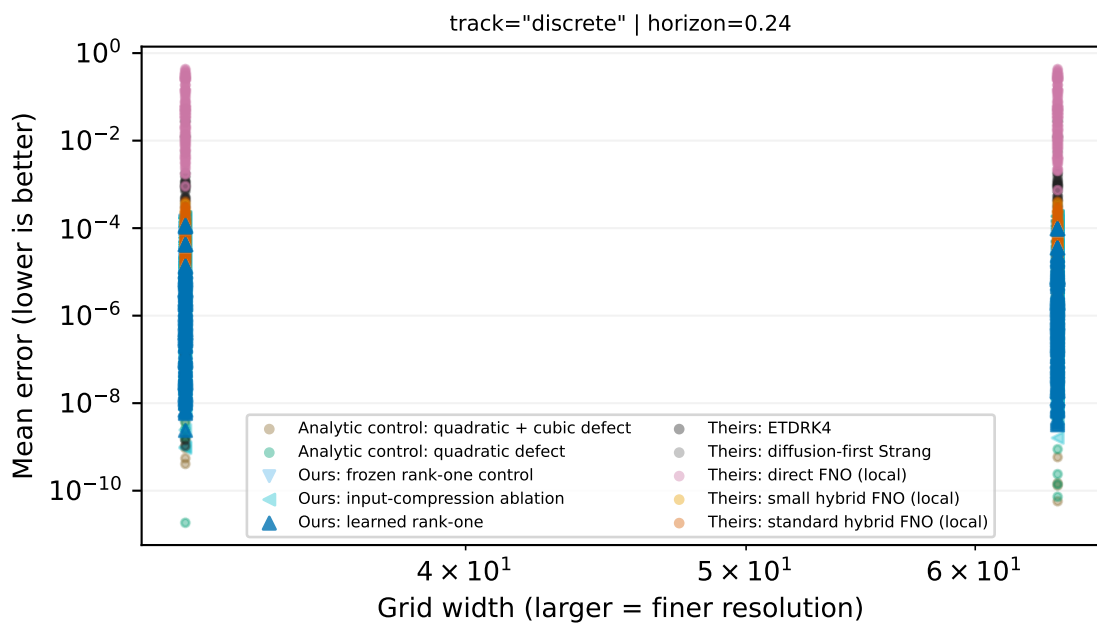
All 6336 finite points; unavailable/nonpositive 0



All 4608 finite points; unavailable/nonpositive 0



All 6336 finite points; unavailable/nonpositive 0



All 4608 finite points; unavailable/nonpositive 0

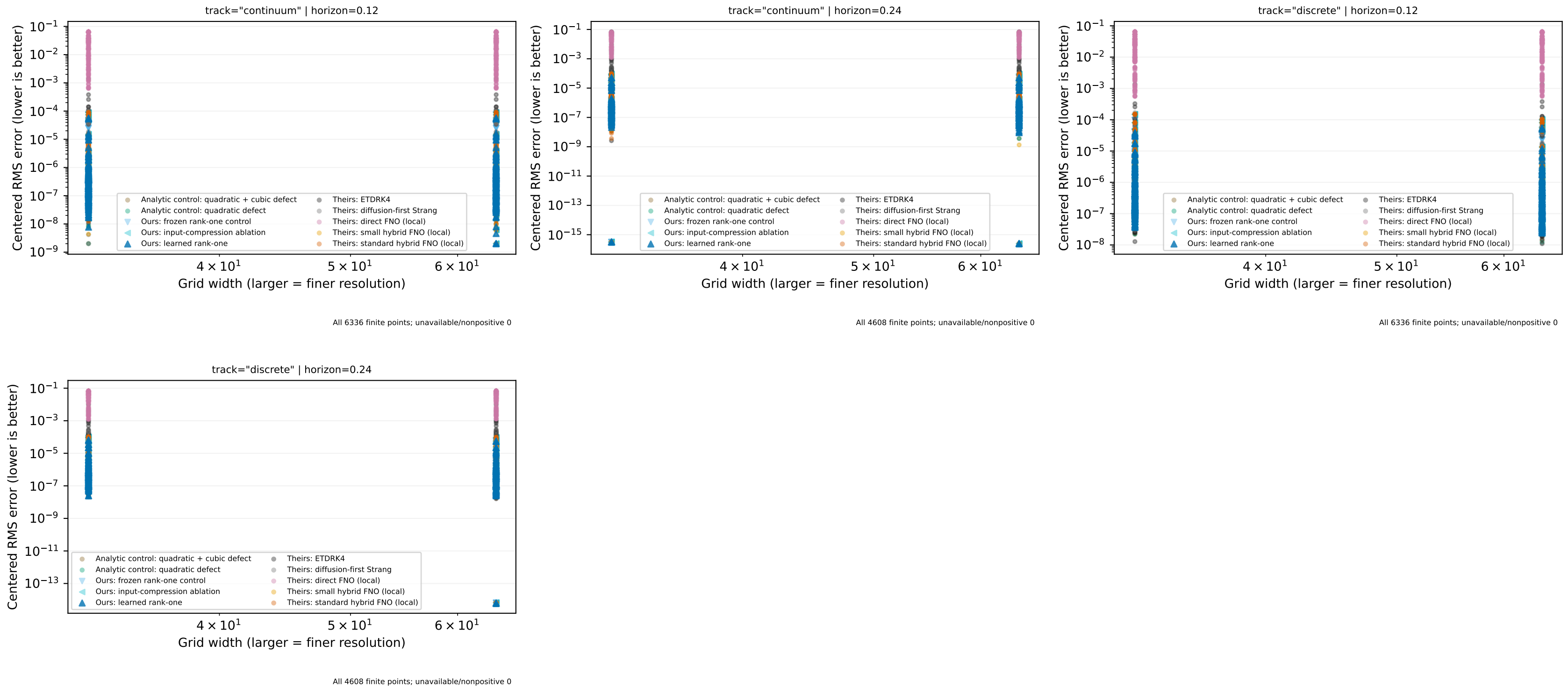
HOW TO READ: Y: lower absolute mean error is better. X: larger grid means finer spatial resolution, not automatically better efficiency. Compare methods at the same grid/target; both axes are logarithmic and exact zeros cannot appear.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Each point retains field, grid, schedule and seed. Spatial and temporal targets are separate; zeros/nonfinite values are counted as unavailable on log axes.

Centered error across spatial resolution



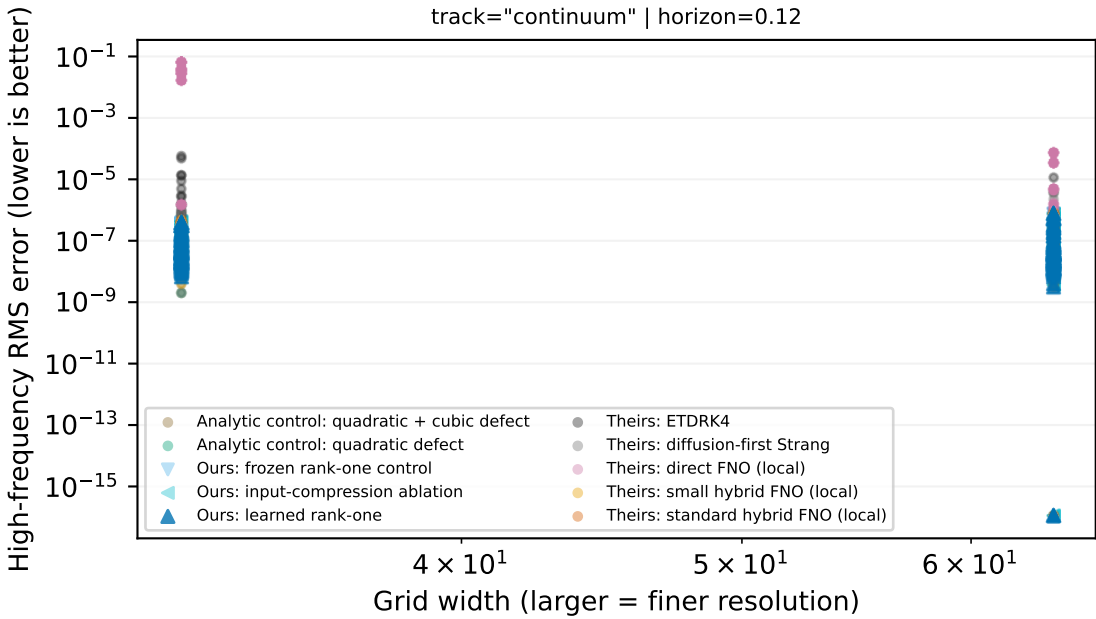
HOW TO READ: Y: lower centered RMS error is better for spatial variation after removing the mean. X: larger grid means finer resolution, not automatically better efficiency. Both axes are logarithmic; this panel does not measure mean accuracy.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

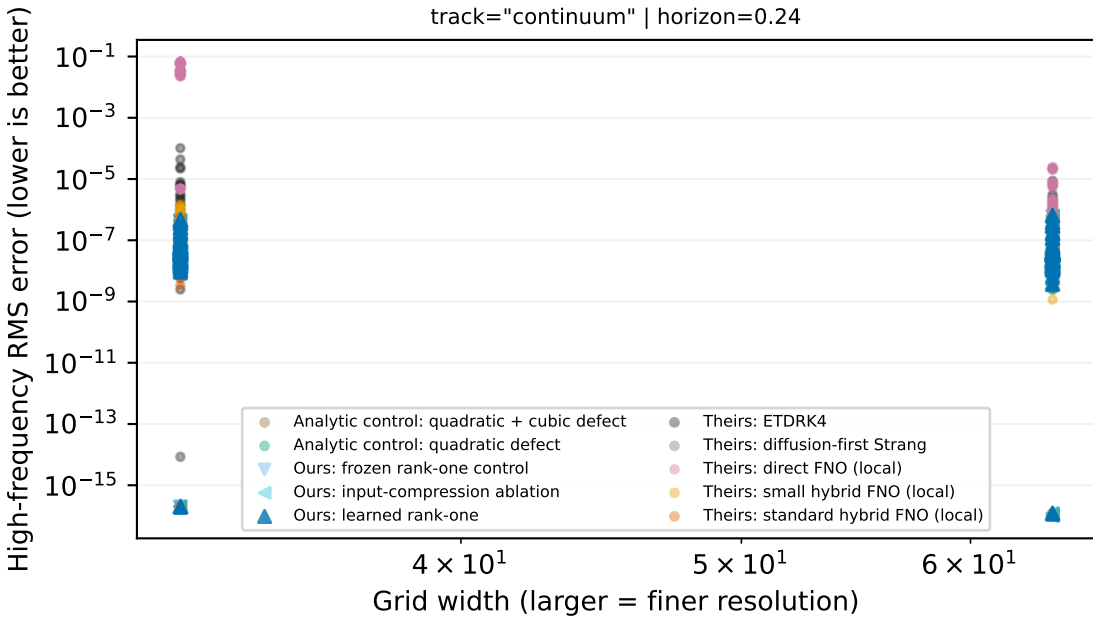
MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Each point retains field, grid, schedule and seed. Spatial and temporal targets are separate; zeros/nonfinite values are counted as unavailable on log axes.

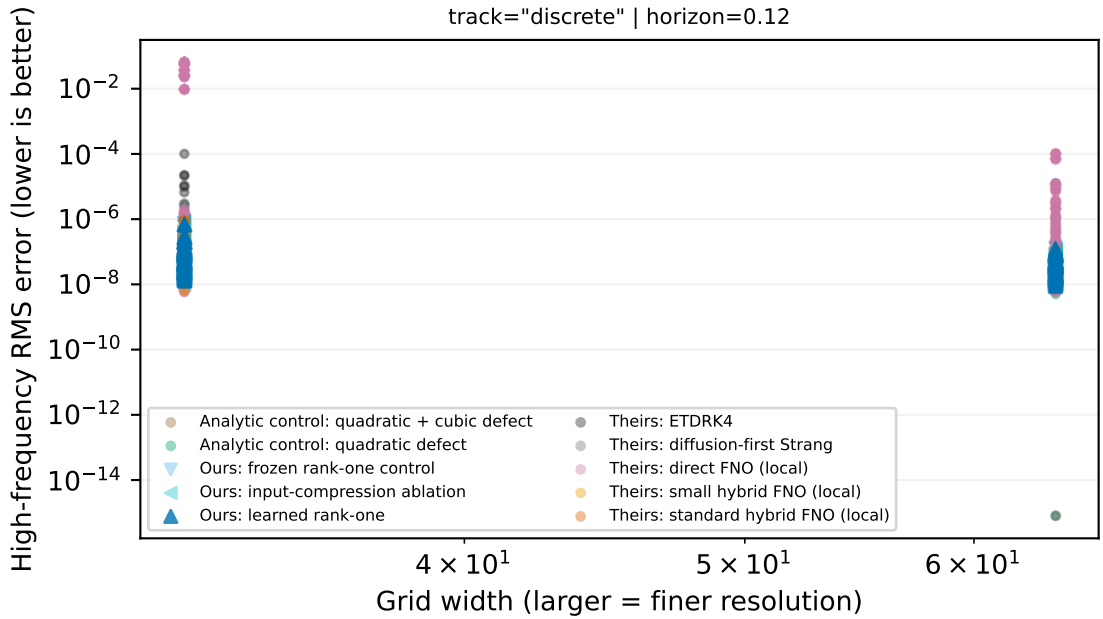
High frequency error across spatial resolution



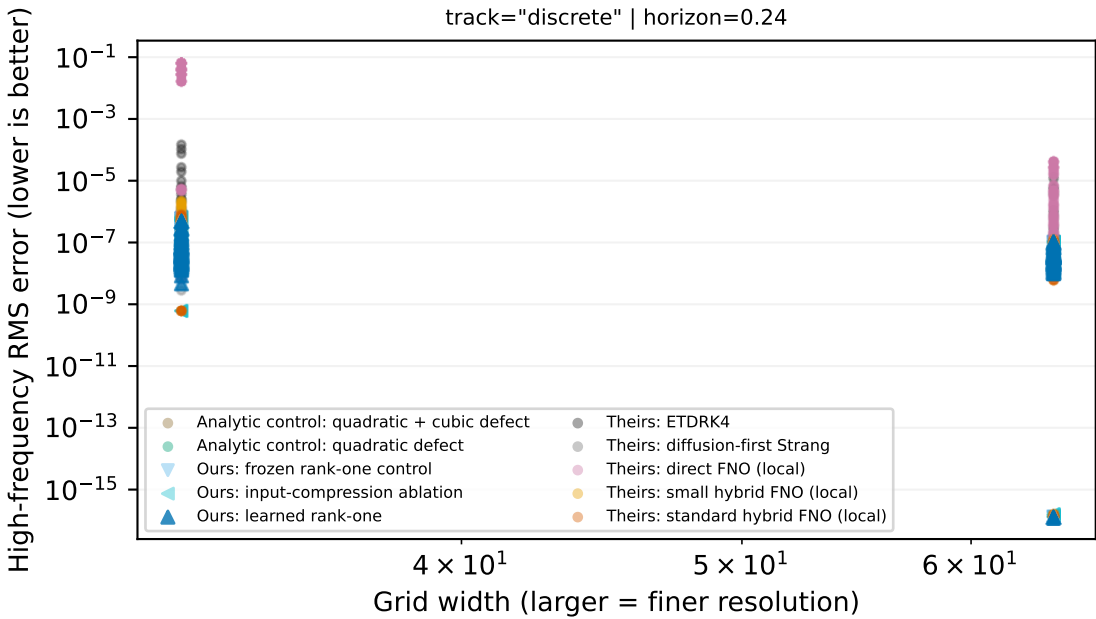
All 6336 finite points; unavailable/nonpositive 0



All 4608 finite points; unavailable/nonpositive 0



All 6336 finite points; unavailable/nonpositive 0



All 4608 finite points; unavailable/nonpositive 0

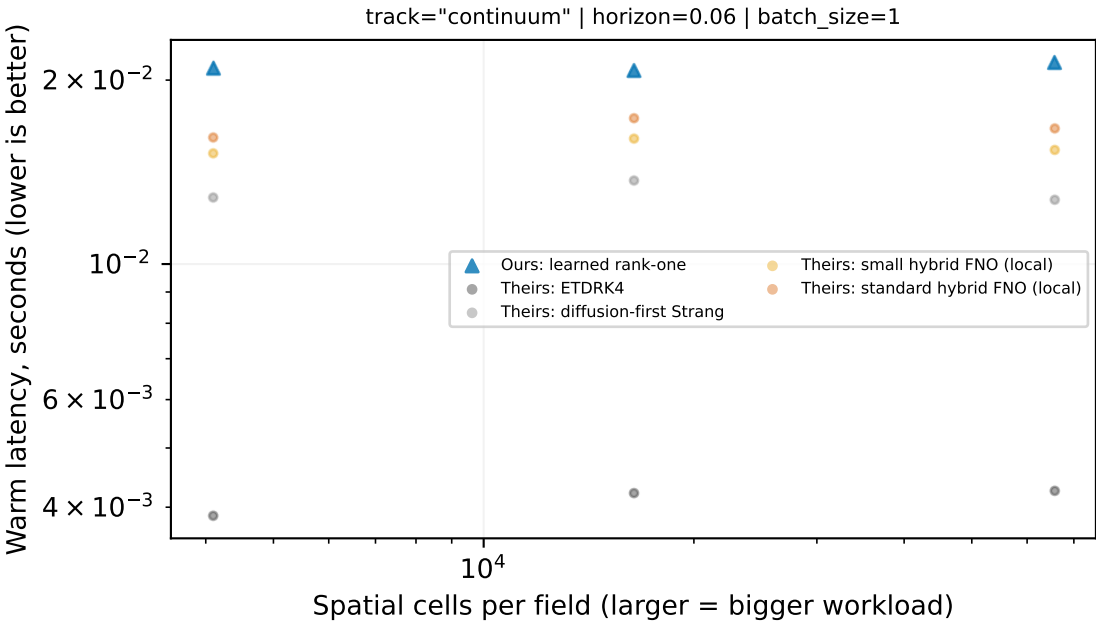
HOW TO READ: Y: lower high-frequency RMS error is better on the declared spectral band. X: larger grid means finer resolution, not a quality score. Compare matched grids/targets; both axes are logarithmic.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

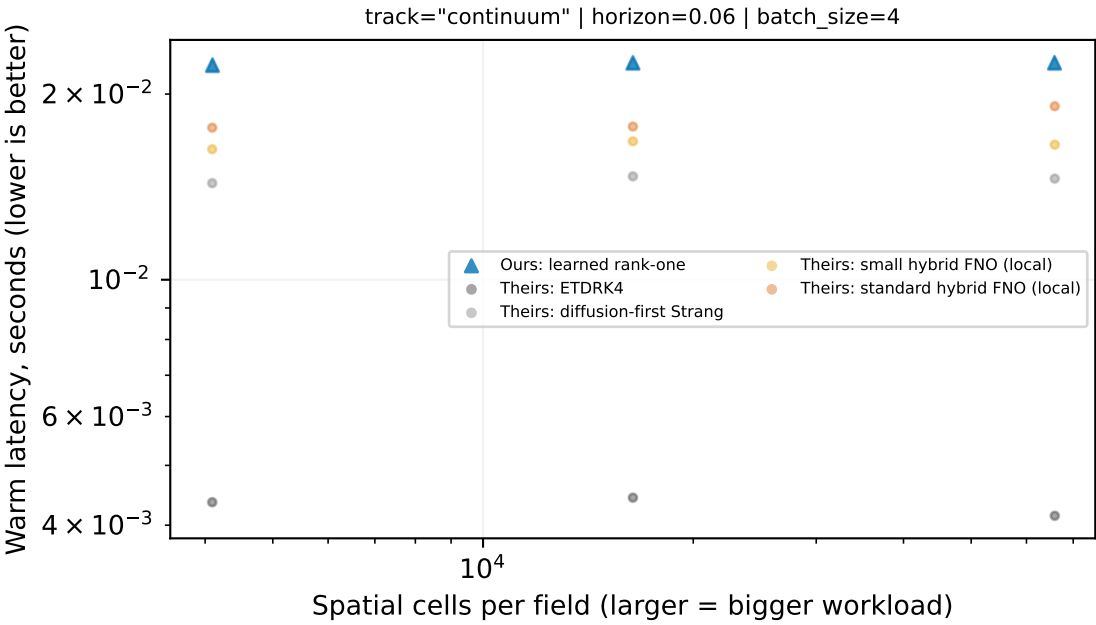
MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Each point retains field, grid, schedule and seed. Spatial and temporal targets are separate; zeros/nonfinite values are counted as unavailable on log axes.

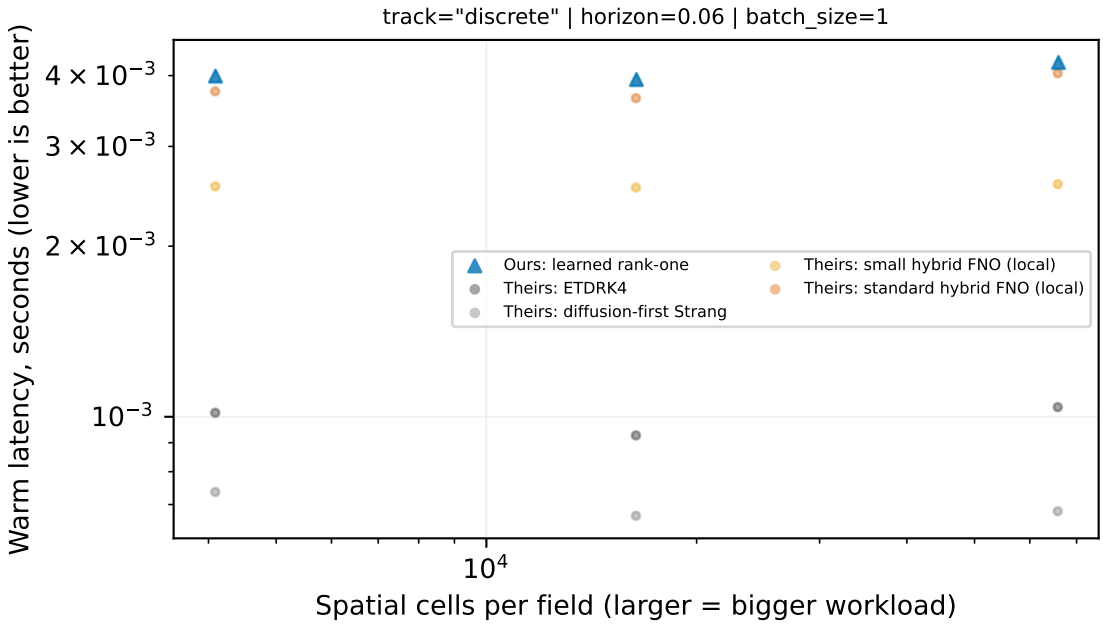
Scaling latency and throughput



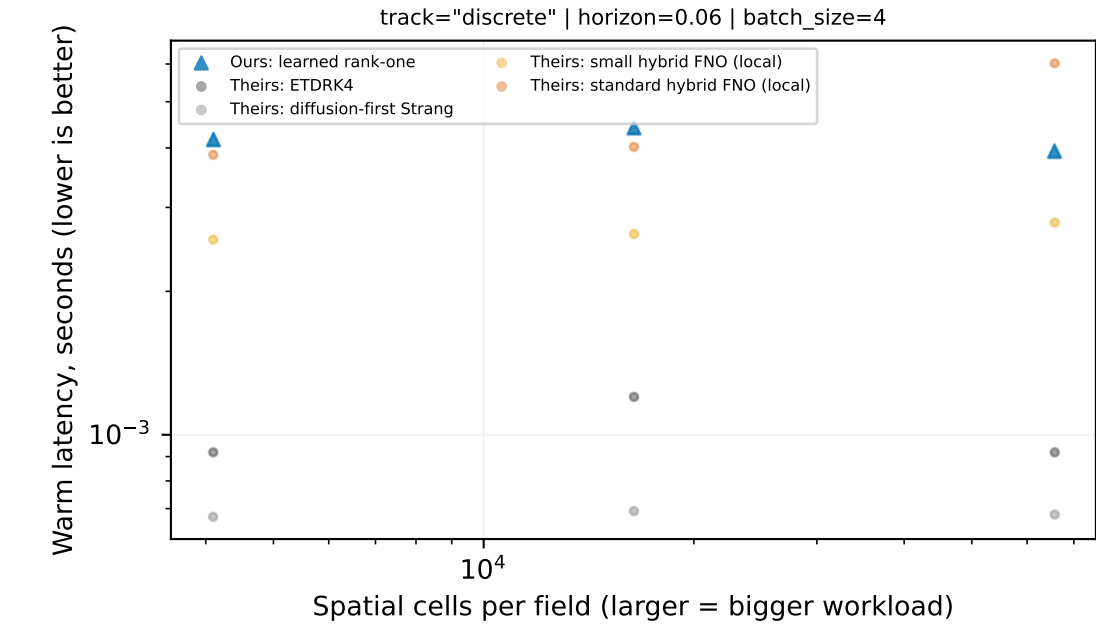
All 15 finite points; unavailable/nonpositive 0



All 15 finite points; unavailable/nonpositive 0



All 15 finite points; unavailable/nonpositive 0



All 15 finite points; unavailable/nonpositive 0

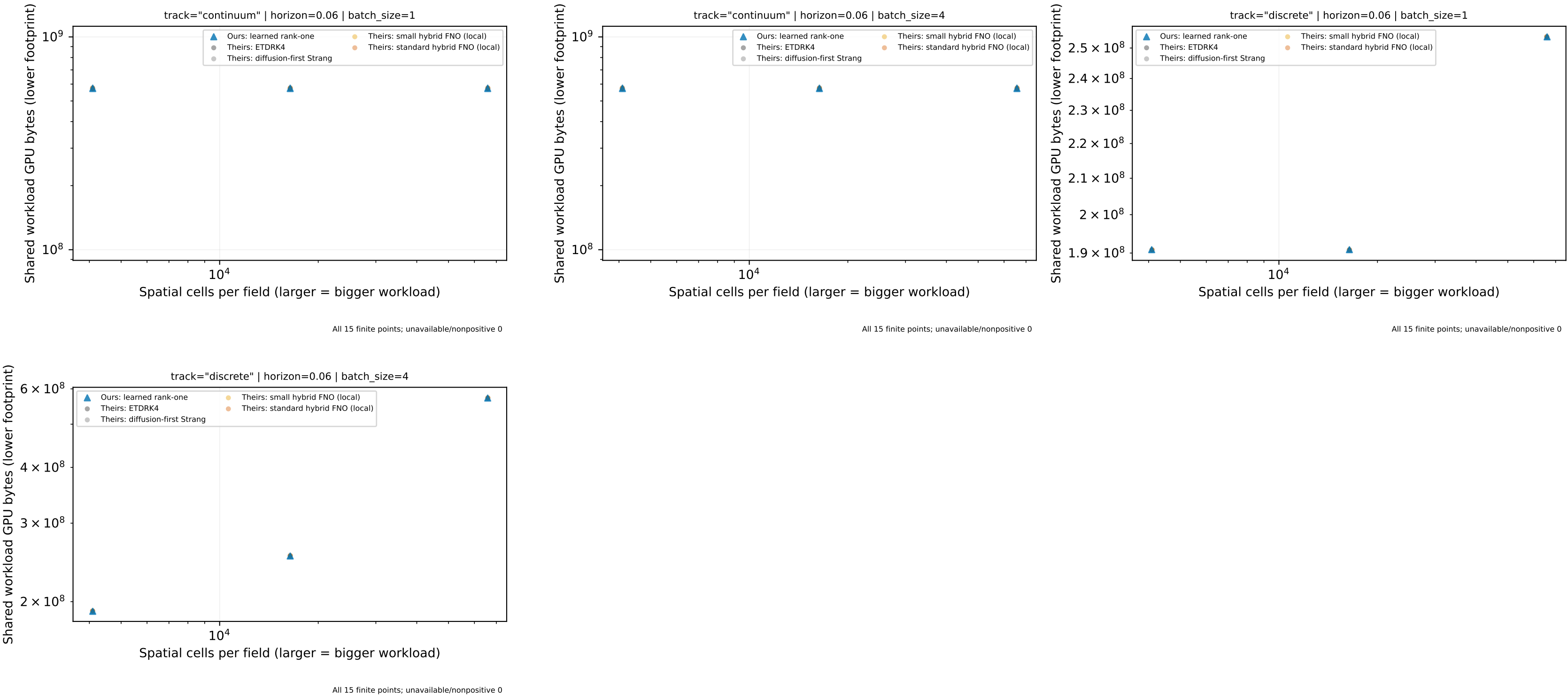
HOW TO READ: Y: lower seconds is faster; at fixed batch size this means higher throughput. X: more cells means a larger problem, not a worse result. Both axes are logarithmic. Compare matched accuracy, grid, batch size, target and final time.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Accuracy is the worst accepted-reference field in each distinct batch; missing references are NA. GPU memory is a shared paired-workload peak, not attributable to a family or shared host RAM.

Scaling peak GPU memory



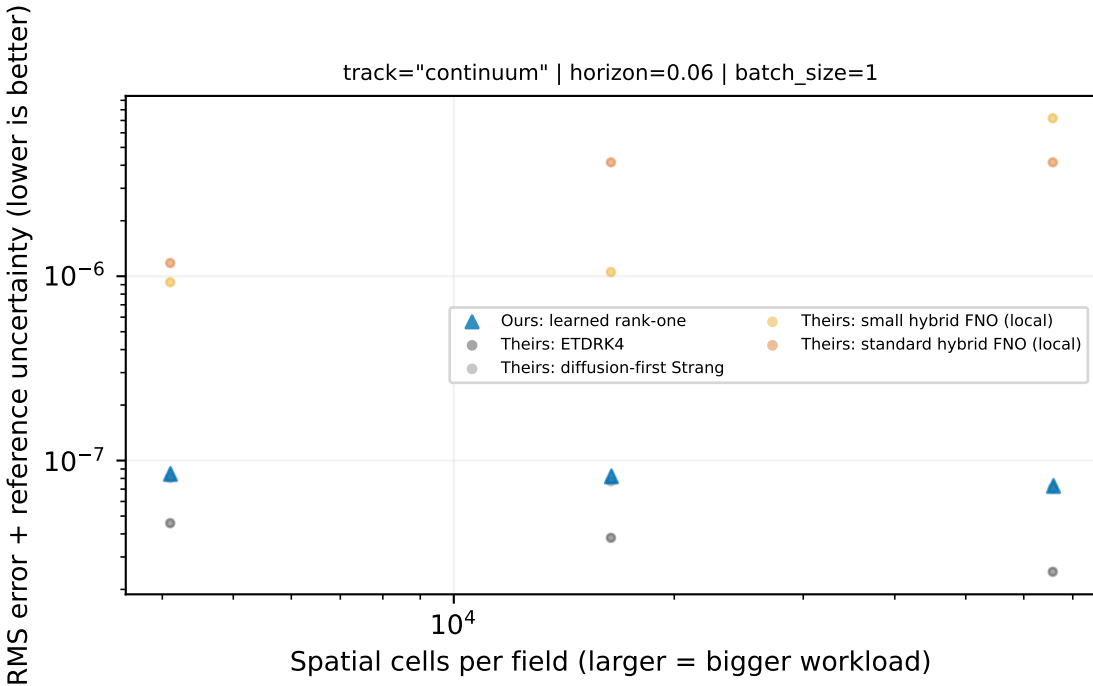
HOW TO READ: Y: lower reserved bytes is less GPU memory for the same workload. This is a shared paired-workload peak, not an isolated method footprint. X is problem size; both axes are logarithmic. Host RAM is not GPU VRAM.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

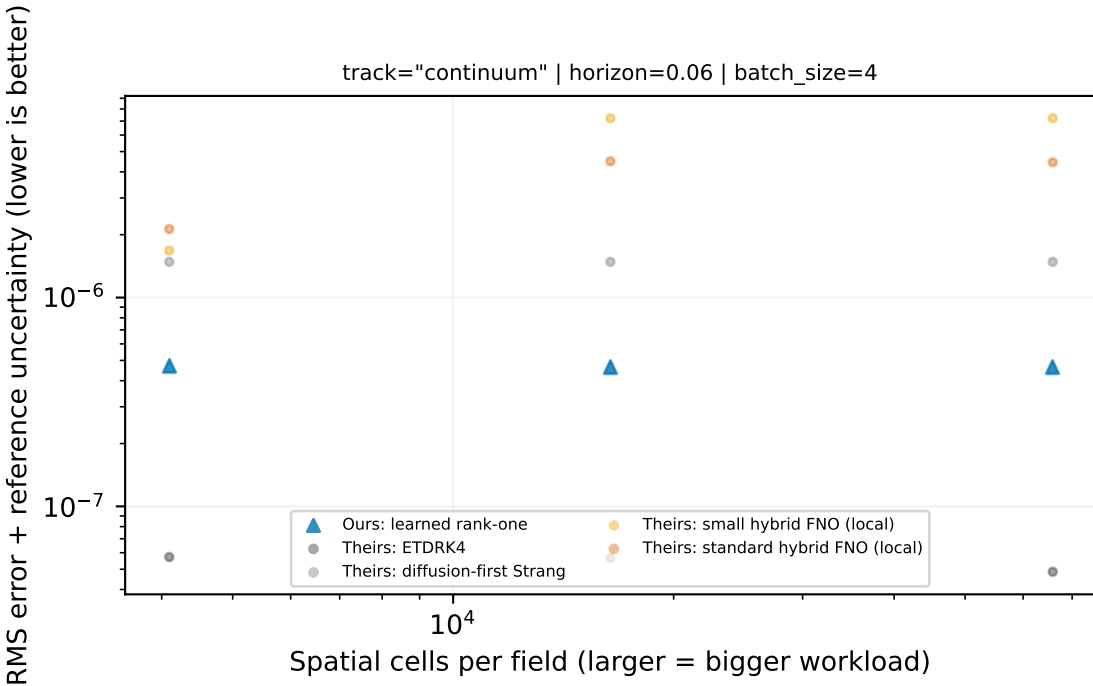
MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Accuracy is the worst accepted-reference field in each distinct batch; missing references are NA. GPU memory is a shared paired-workload peak, not attributable to a family or shared host RAM.

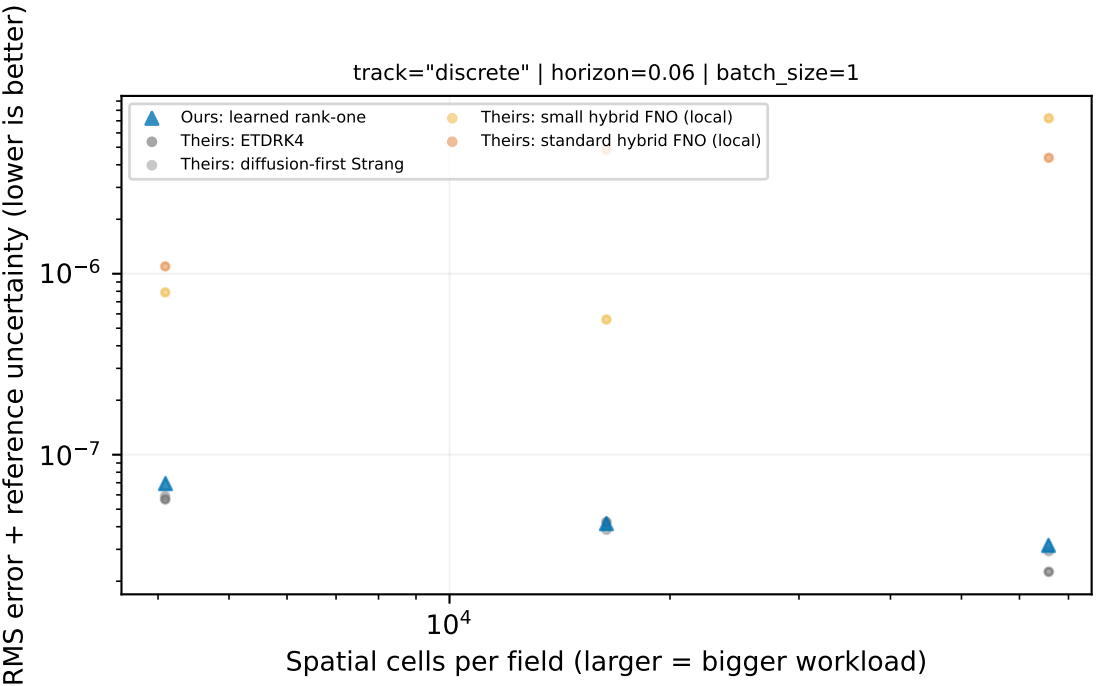
Scaling accepted reference accuracy



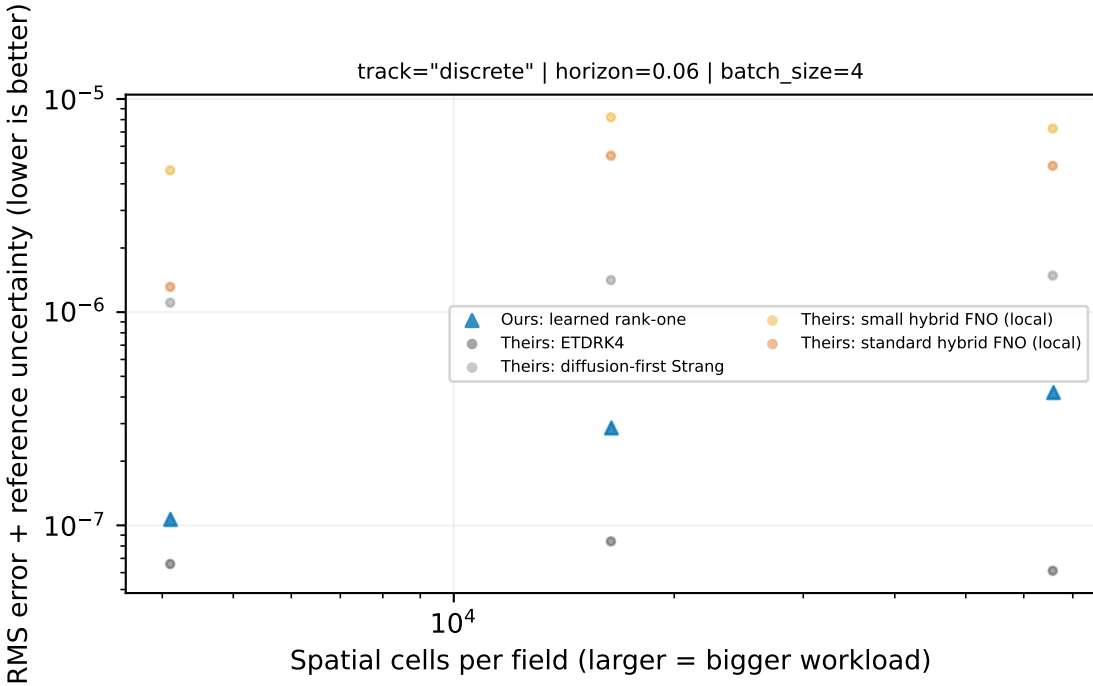
All 15 finite points; unavailable/nonpositive 0



All 15 finite points; unavailable/nonpositive 0



All 15 finite points; unavailable/nonpositive 0



All 15 finite points; unavailable/nonpositive 0

HOW TO READ: Y: lower worst-field RMS plus accepted-reference uncertainty is better. X: more cells means a larger problem; compare the same workload. Both axes are logarithmic. Missing accepted references remain NA, not accurate.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Accuracy is the worst accepted-reference field in each distinct batch; missing references are NA. GPU memory is a shared paired-workload peak, not attributable to a family or shared host RAM.

Deployment complete cost components

track=NA | final_time=NA | mode=NA

NA — no verified measurements for this panel

HOW TO READ: Y: lower complete runtime is better at matched accuracy. X lists different cost scopes. Proposal/estimator/fallback components sum to attributed time; do not add that total to complete-call time. Blocked deployment is NA, not free computation.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: First three components sum to attributed_seconds for a representative run; cost_seconds is independent complete warmed median. Do not sum these totals together. Missing or blocked deployment remains NA.

Policy coverage and false acceptance

track=NA | final_time=NA | mode=NA

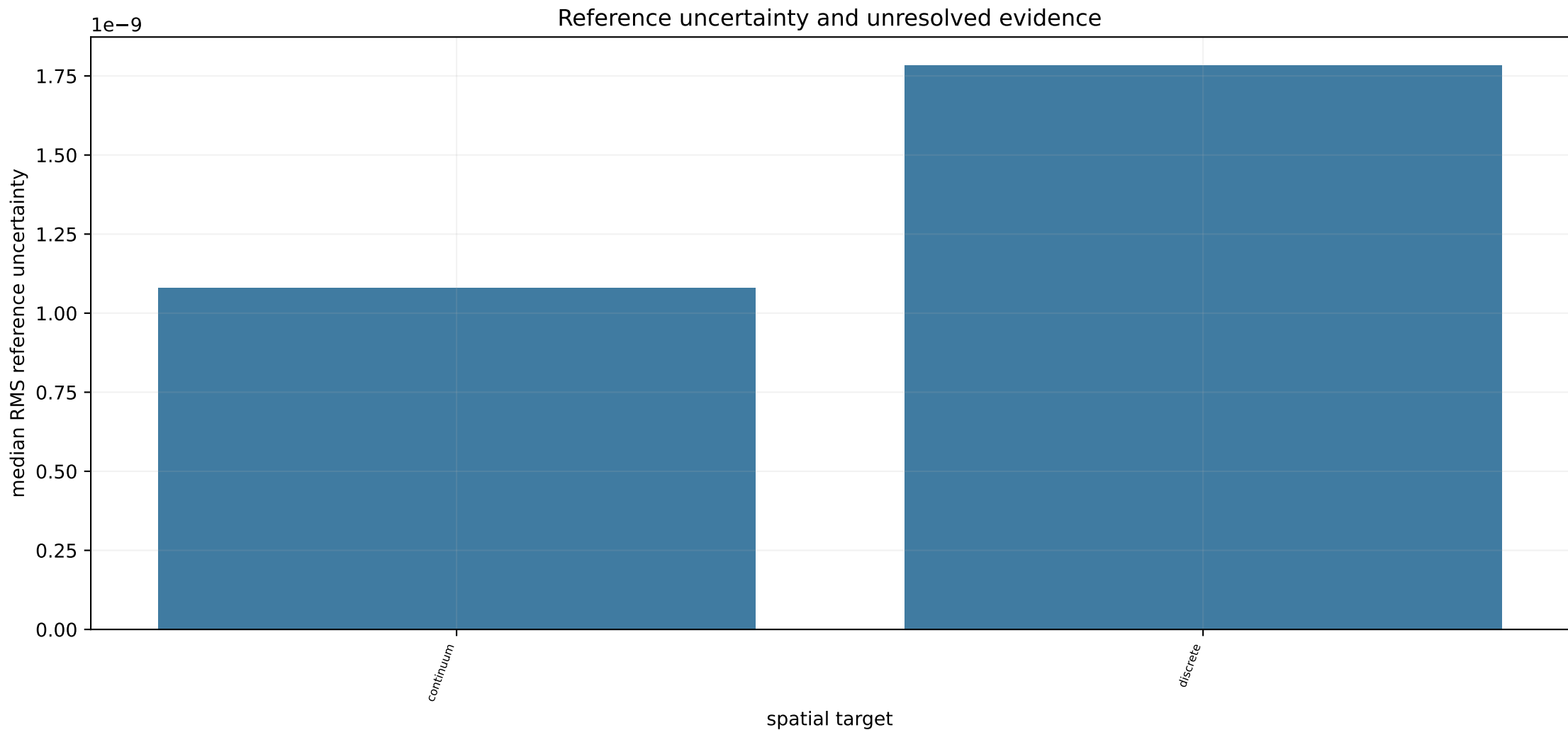
NA — no verified measurements for this panel

HOW TO READ: Fewer false acceptances is better. More accepted queries is useful only with verified accuracy and cost savings; fallback can be the correct decision. Y counts decisions, not independent fields or a risk guarantee.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Decisions use calibration and deployable features only. Confirmation truth audits decisions afterward; finite empirical coverage is not a distribution-free guarantee.



HOW TO READ: Y: lower uncertainty permits finer accuracy distinctions, but does not prove the reference is exact. X separates spatial targets, not a ranking.

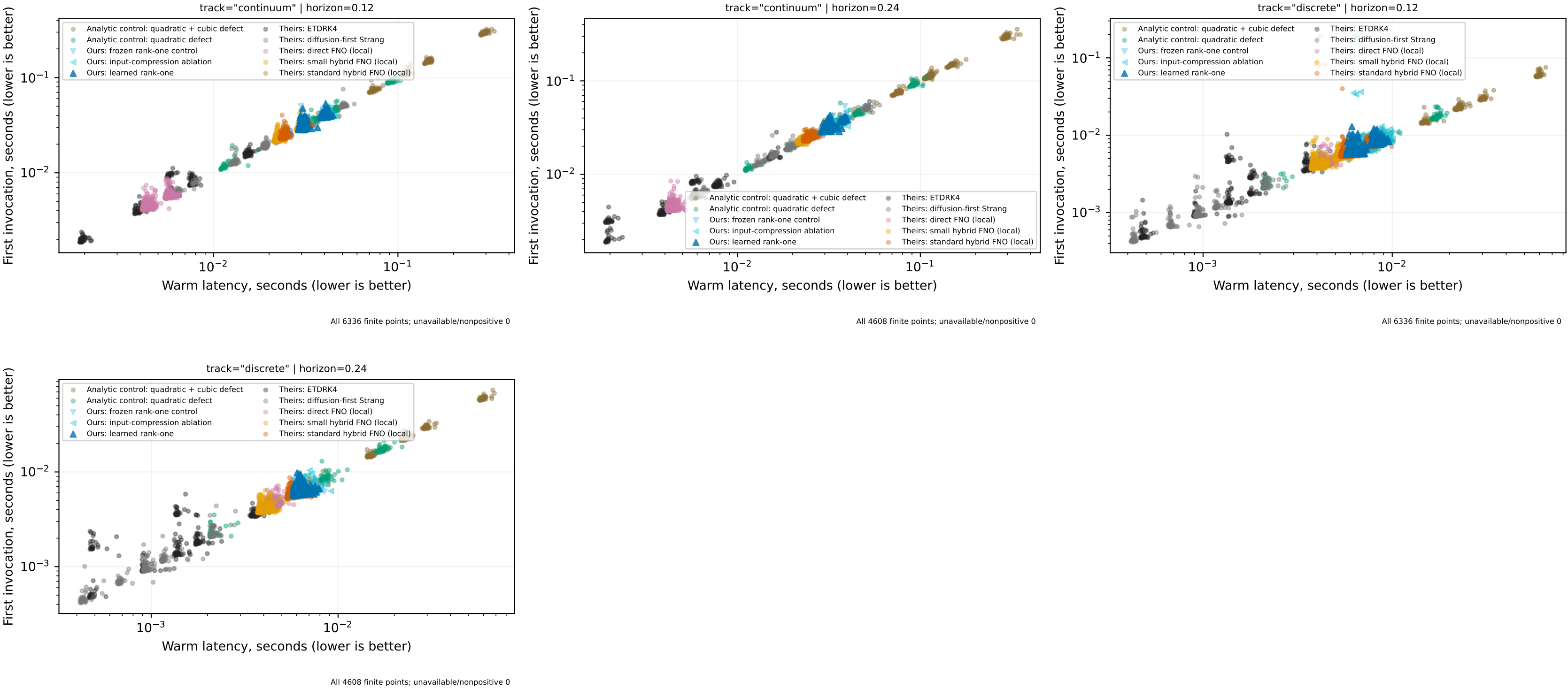
Differences beneath uncertainty are unresolved; missing evidence is not zero.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: A tighter solver error beneath reference uncertainty is unresolved. Spatial and time refinement are not interchangeable.

First invocation versus randomized warmed timing



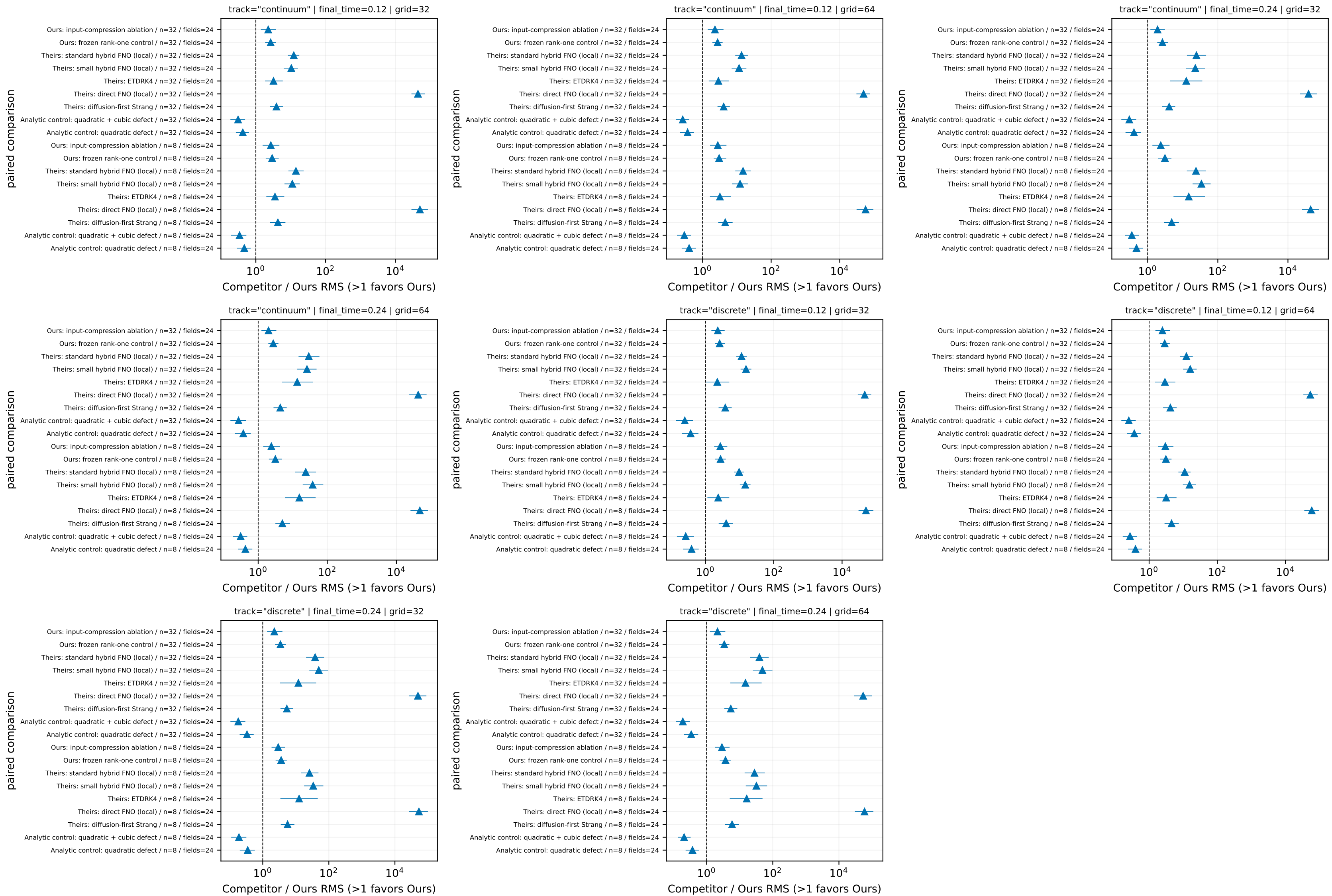
HOW TO READ: Lower-left means lower time in both scopes: X warmed median, Y first invocation. Both axes are logarithmic; compare identical workloads and accuracy. First invocation is not guaranteed process-cold startup or an amortization result.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: First observed invocation is not guaranteed process-cold startup. Speed comparisons use randomized warmed measurements; first-call savings do not establish amortization.

Independent field intervals error_ratio



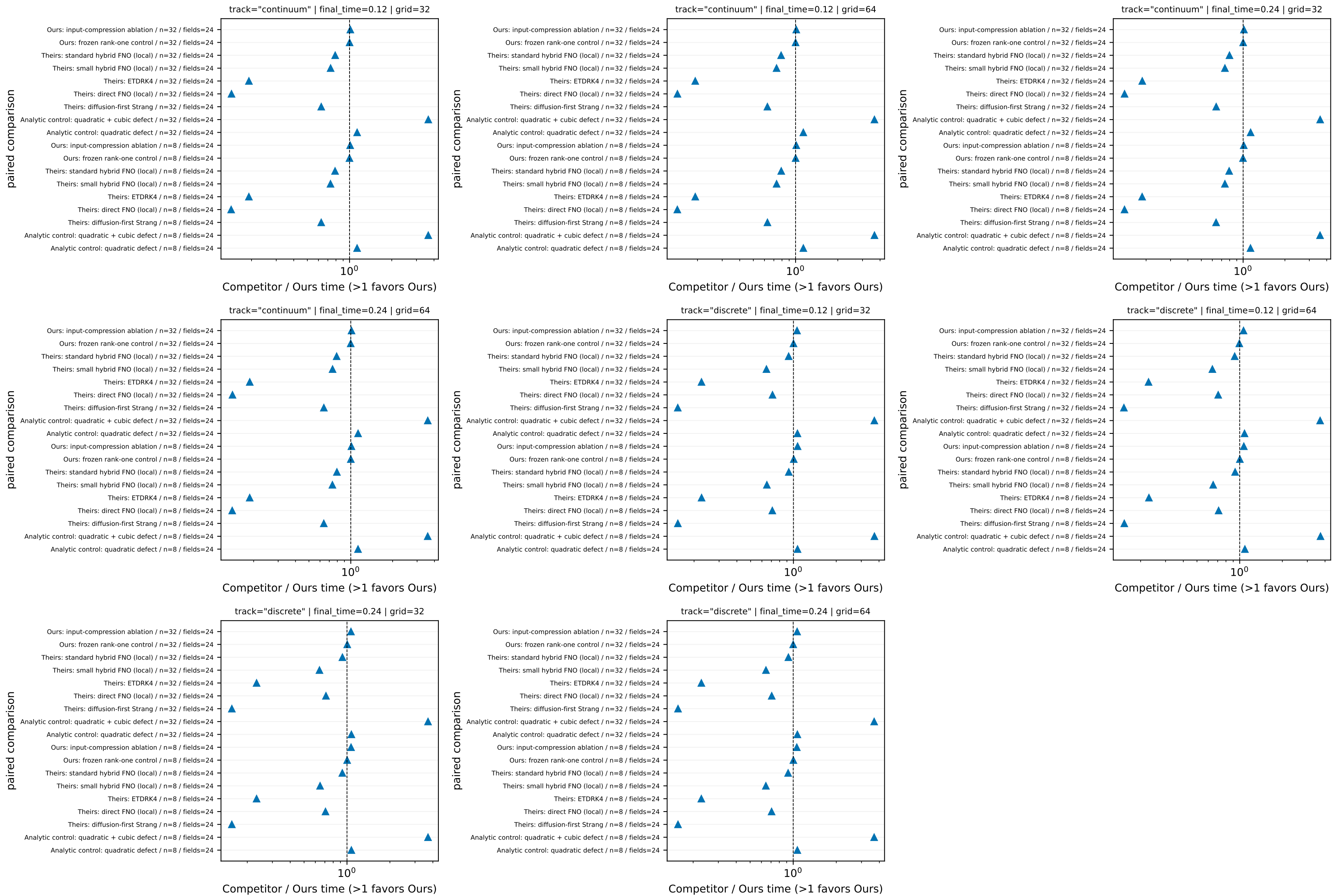
HOW TO READ: X = competitor RMS / Ours learned rank-one RMS: right of 1 favors Ours; left of 1 favors the named competitor. Higher is better for Ours; 1 is a tie. Logarithmic X; an interval crossing 1 leaves the error direction unresolved.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Descriptive field-cluster bootstrap; repeated seeds and physics do not increase independent field count. Missing intervals remain absent, never zero width.

Independent field intervals speed_ratio



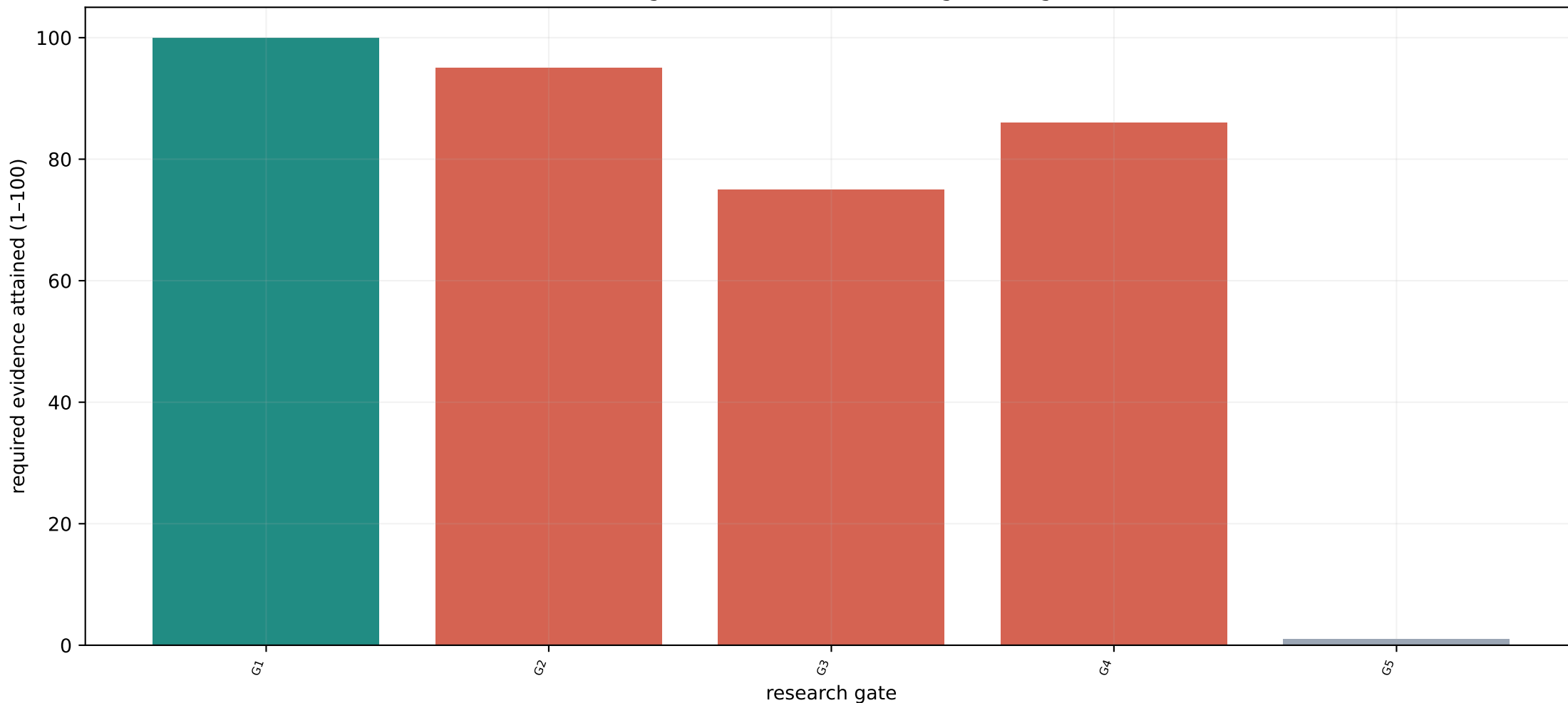
HOW TO READ: X = competitor time / Ours learned rank-one time: right of 1 favors Ours; left of 1 favors the named competitor. Higher is faster for Ours; 1 is a tie. Logarithmic X; a speed ratio alone does not establish matched-accuracy utility.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Descriptive field-cluster bootstrap; repeated seeds and physics do not increase independent field count. Missing intervals remain absent, never zero width.

Five gate evidence and missing coverage



HOW TO READ: Y: higher scores mean more declared evidence attained, not better model quality. Teal GOOD, coral BAD and gray NA retain the gate verdict; a high NA score is not success. X lists research gates, not a performance ranking.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Teal GOOD, coral BAD, gray NA. Scores are predeclared check-weight attainment, never scientific quality, effect size, or a proof.

Independent function risk and available evidence

track=NA | mode=NA

NA — no verified measurements for this panel

HOW TO READ: Y: lower empirical risk and a lower valid upper bound are better. The target is a requirement, not an observed result; compare the bound with that target. NA or no accepted fields means unestablished risk, not zero risk.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Binomial upper bound is conditional on iid comparable fields and a frozen policy. No accepted functions means NA; repeated query rows are not independent trials.

Supported amortization including offline costs

track=NA

Unestablished:

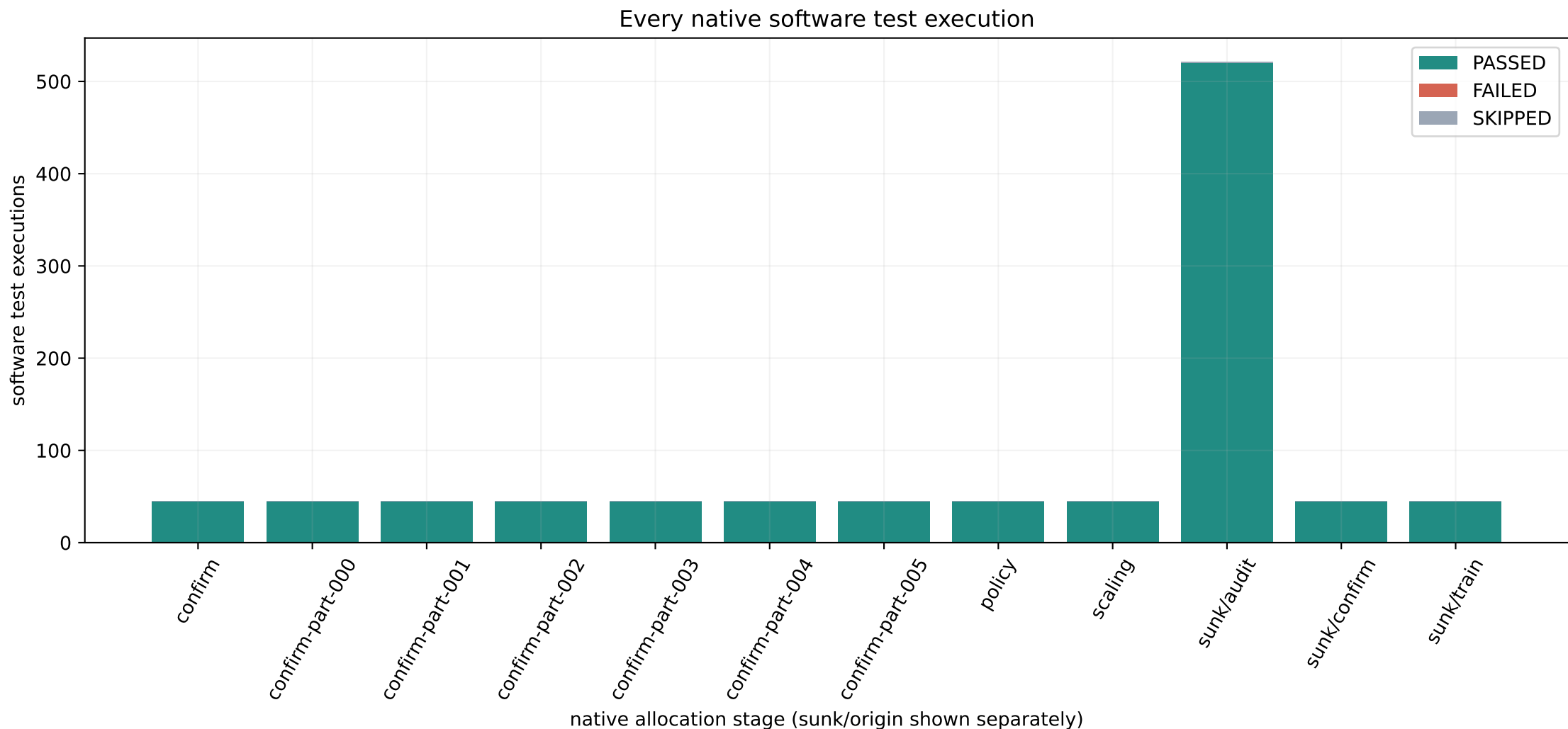
NA — no verified measurements for this panel

HOW TO READ: Y: fewer break-even queries is better when matched accuracy and positive savings are established. X lists policy modes. NA means amortization was not established; it must not be read as zero required queries.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Requires matched accuracy, accepted learned queries and positive saving uncertainty bound. All-classical routing or startup timing differences cannot earn neural amortization.



HOW TO READ: Teal PASSED is favorable readiness evidence; coral FAILED needs attention; gray SKIPPED provides no pass evidence. Y counts executions, not model quality. Repeated passes across allocations do not add independent scientific evidence.

METHODS: Ours = proposed rank-one family and its controls. Theirs = established-method comparators implemented here; local FNO variants are not published-paper reproductions. Analytic control = untrained attribution reference, not an external FNO result or an ownership/novelty claim.

MARKERS: Blue solid line / up triangle = Ours, learned rank-one; light-blue dotted / down triangle = frozen; cyan dashed / left triangle = ablation. Other methods use thin patterned lines / circles. Method colors are consistent; ratios use candidate styling. Hatched bars are controls. Learning bands show observed low/high ranges with faint fill and thin opaque borders. Heatmap colors show accuracy; partial-diagnostic colors show evidence status.

SCOPE: Repeated cases in different allocations remain separate executions. Skipped tests never count as passed GPU readiness; software correctness is not scientific superiority.